

- 1 Contest
- 2 Data structures
- 3 Numerical
- 4 Number theory
- 5 Combinatorial
- 6 Graph
- 7 Geometry
- 8 Strings
- 9 Various

Contest (1)

setup

5 lines

```
create d.sh & code.cpp in foo.
for i in {A..J}; do cp -r foo $i; done;
for i in {A..J}; do chmod +x $i/d; done;
create debug.h in foo.
for i in {A..J}; do cp foo/debug.h $i; done;
```

d.sh

14 lines

```
#!/usr/bin/env bash
if ! g++ -std=c++23 -DLOCAL -Wshadow -Wall -g \
  -fsanitize=address -fsanitize=undefined -D_GLIBCXX_DEBUG \
  -fmax-errors=2 code.cpp -o a.out
then
  exit 1
fi
echo
echo Input:
cat $1
echo
echo Output:
./a.out < $1
echo
```

run.sh

11 lines

```
#!/bin/bash
for inp in in in in[0-9]*; do
  if [ -f $inp ]; then
    echo
    echo "Input: $inp"
    cat $inp
    echo Output:
    ./a.out < $inp
    echo
  fi
done
```

f.sh

1 lines

```
g++ -std=c++17 -O2 -Wno-unused-result code.cpp -o a.out
```

template.cpp

56 lines

```
#include <bits/stdc++.h>
using namespace std;

// *st.find_by_order(index) = value at index
// st.order_of_key(x) = no. of elements strictly less than x
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
#define ordered_set tree<int, null_type,less<int>, \
  rb_tree_tag,tree_order_statistics_node_update>

#define int long long
#define ll long long

#define all(v) v.begin(),v.end()
#define range(v, i, j) v.begin()+i, v.begin()+j+1
#define rep(i, a, b) for(long long i = (a); i < (b); ++(i))
#define L(i, a, b) for(long long i = (a); i <= (b); ++(i))
#define R(i, a, b) for(long long i = (a); i >= (b); --(i))
#define sz(x) (ll)(x.size())
#define extract(m, x) \
{ auto it = (m).find(x); if (it != (m).end()) (m).erase(it); }
#define gp " "
#define nl "\n"

#define isSet(x, i) ((x>>i)&1)
#define setbit(x, i) (x | (1LL<<i))
#define resetbit(x, i) (x & ~(1LL << i))
#define toggleBit(x, i) ((x) ^ (1LL << (i)))
#define clz(x) __builtin_clzll(x)
#define ctz(x) __builtin_ctzll(x)
#define csb(x) __builtin_popcountll(x)
#define msb(x) ((x) ? (63 - __builtin_clzll((ll)(x))) : -1)
#define lsb(x) ((x) ? (__builtin_ctzll((ll)(x))) : -1)

#ifdef LOCAL
#include "debug.h"
#else
#define deb(...)
#endif

mt19937_64 \
rnd(chrono::steady_clock::now().time_since_epoch().count());
const ll inf = 1e18;

void prep(){

void solve(int tcase){

int32_t main() {
  ios_base::sync_with_stdio(false); cin.tie(NULL);
  prep();
  int t = 1;
  // cin >> t;
  L(i, 1, t) solve(i);
}

debug.h
```

debug.h

46 lines

```
#pragma once
template<class T>
auto pr(T x) -> decltype(cerr<<x, void()) {cerr<<x;}
void pr(string s) {cerr << "'" << s << "'";}
template<class A, class B>
void pr(pair<A,B> p){
  cerr << "{"; pr(p.F); cerr << ", ";
  pr(p.S); cerr << "}";
}
```

```
template<class... A>
void pr(tuple<A...> t){
  cerr << "(";
  bool f = true;
  apply([&](auto... x){
    ((cerr << (f ? (f=false, "") : ", "), pr(x)), ...);
  }, t);
  cerr << ")";
}
```

```
template<class T>
auto pr(T v) -> decltype(v.begin(), void()){
  cerr << "[";
  bool f = 1;
  for(auto x : v){
    if(!f) cerr << ", ";
    f = false;
    pr(x);
  }
  cerr << "]";
}
```

```
void d_b(const char* s) {}
```

```
template<class T, class... U>
void d_b(const char* s, T t, U... u) {
  while (*s == ',' || *s == ' ') s++;
  const char* c = strchr(s, ',');
  int len = c ? c - s : strlen(s);
  cerr.write(s, len) << " :";
  pr(t);
  if (sizeof...(u)) { cerr << " "; d_b(c, u...); }
  else { cerr << endl; }
}
#define deb(...) \
  cerr << __LINE__ << "| ", d_b(#__VA_ARGS__, __VA_ARGS__)
```

stress.sh

16 lines

```
set -e
g++ gen.cpp -o gen
g++ brute.cpp -o bru
for((i = 1; ; ++i)); do
  ./gen $i > tcase
  ./a.out < tcase > myans
  ./bru < tcase > corans
  diff -Z myans corans > /dev/null || break
  echo "Passed test: " $i
done
echo "WA on test:"
cat tcase
echo "Your answer:"
cat myans
echo "Correct answer:"
cat corans
```

gen.cpp

4 lines

```
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().
  count());
inline ll rnd(ll l, ll r) {
  return uniform_int_distribution<ll>(l, r)(rng);
}
```

brute.cpp

1 lines

```
// Write brute force for stress testing
```

```
8 lines
@echo off /// set SRC=code.cpp /// set EXE=a
g++ -std=c++17 -DLOCAL -Wshadow -Wall -g -D_GLIBCXX_DEBUG -fmax
    -errors=2 %SRC% -o %EXE%
if errorlevel 1 (
    echo Compilation failed! /// exit /b
)
echo. /// echo Input: /// type %1 /// echo.
echo. /// echo Output: /// %EXE% < %1 /// echo.
/// alt : gnu++14 or ++11 or ++1z
```

```
48 lines
#include "euclid.h"
struct mint{
    ll x; mint() { x=0; }
    mint(ll xx) {x = xx%mod; if(x<0)x+=mod; }

    mint& operator+=(mint b){ x=(x + b.x)%mod; return *this; }
    mint& operator*=(mint b){ x=(x * b.x)%mod; return *this; }
    mint& operator-=(mint b){ x=(x - b.x +mod)%mod; return *this; }
    mint& operator/=(mint b){ return *this *= b.inv(); }

    friend mint operator+(mint a,mint b){ return a+=b; }
    friend mint operator-(mint a,mint b){ return a-=b; }
    friend mint operator*(mint a,mint b){ return a*=b; }
    friend mint operator/(mint a,mint b){ return a/=b; }

    mint operator-() { return mint()-*this; }
    mint inv() { return power(mod-2); } // prime mod

    // mint inv() { // any mod
    //     ll x, y, g = euclid(a.x, mod, x, y);
    //     assert(g == 1); return mint((x + mod) % mod);
    // }

    mint power(ll n) {
        mint a=*this, res=1;
        while(n) { if(n&1)res*=a; a*=a; n>>=1;}
        return res;
    }

    explicit operator bool() const { return x != 0; }
    friend ostream& operator<<(ostream& os, const mint& m) \
    { return os << m.x; }
};

mint power(mint a,ll n){ return a.power(n); }

const int N=2000005; mint fact[N],inv_fact[N];
void prep_factorials(){
    fact[0] = 1;
    for(int i=1; i<N; ++i) fact[i] = fact[i-1] * i;
    inv_fact[N-1] = fact[N-1].inv();
    for(int i=N-2; i>=0; --i) inv_fact[i] = inv_fact[i+1]*(i+1);
}

mint nCr(int n,int r){
    if(r<0 || r>n) return mint();
    return fact[n] * inv_fact[r] * inv_fact[n-r];
}
```

1.1 Recurrences

If $a_n = c_1a_{n-1} + \dots + c_ka_{n-k}$, and $r_1, \dots, r_k$ are distinct roots of $x^k - c_1x^{k-1} - \dots - c_k$, there are $d_1, \dots, d_k$ s.t.

$$a_n = d_1r_1^n + \dots + d_kr_k^n.$$

Non-distinct roots r become polynomial factors, e.g.

$$a_n = (d_1n + d_2)r^n.$$

1.2 Trigonometry

$$\tan(v+w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$
$$\sin v + \sin w = 2 \sin \frac{v+w}{2} \cos \frac{v-w}{2}$$
$$\cos v + \cos w = 2 \cos \frac{v+w}{2} \cos \frac{v-w}{2}$$

$$(V+W)\tan(v-w)/2 = (V-W)\tan(v+w)/2$$

where V, W are lengths of sides opposite angles v, w .

$$a \cos x + b \sin x = r \cos(x - \phi)$$
$$a \sin x + b \cos x = r \sin(x + \phi)$$

where $r = \sqrt{a^2 + b^2}, \phi = \text{atan2}(b, a)$.

1.3 Geometry

1.3.1 Triangles

Side lengths: a, b, c

Semiperimeter: $p = \frac{a+b+c}{2}$

Area: $A = \sqrt{p(p-a)(p-b)(p-c)}$

Circumradius: $R = \frac{abc}{4A}$

Inradius: $r = \frac{A}{p}$

Length of median (divides triangle into two equal-area triangles):

$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

Length of bisector (divides angles in two):

$$s_a = \sqrt{bc \left[1 - \left(\frac{a}{b+c} \right)^2 \right]}$$

Law of sines: $\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R}$

Law of cosines: $a^2 = b^2 + c^2 - 2bc \cos \alpha$

Law of tangents: $\frac{a+b}{a-b} = \frac{\tan \frac{\alpha+\beta}{2}}{\tan \frac{\alpha-\beta}{2}}$

1.3.2 Quadrilaterals

With side lengths a, b, c, d , diagonals e, f , diagonals angle θ , area

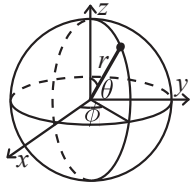
A and magic flux $F = b^2 + d^2 - a^2 - c^2$:

$$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is 180° ,

$ef = ac + bd$, and $A = \sqrt{(p-a)(p-b)(p-c)(p-d)}$.

1.3.3 Spherical coordinates



$$x = r \sin \theta \cos \phi \quad r = \sqrt{x^2 + y^2 + z^2}$$
$$y = r \sin \theta \sin \phi \quad \theta = \text{acos}(z/\sqrt{x^2 + y^2 + z^2})$$
$$z = r \cos \theta \quad \phi = \text{atan2}(y, x)$$

1.4 Sums

$$c^a + c^{a+1} + \dots + c^b = \frac{c^{b+1} - c^a}{c - 1}, c \neq 1$$

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(2n+1)(n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

1.5 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, (-\infty < x < \infty)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, (-\infty < x < \infty)$$

Data structures (2)

OrderedMultiset.h

Description: ...

Time: ...

93fcc8, 22 lines

```
template <typename T>
struct ordered_multiset {
    using Tpair = pair<T, int>;
    tree<Tpair, null_type, less<Tpair>, rb_tree_tag \
    tree_order_statistics_node_update> t;
    int _idx = 0;
    void insert(const T &x) { t.insert({x, _idx++}); }
    void erase(const T &x) {
```

```

    auto it = t.lower_bound({x, 0});
    if (it != t.end() && it->first == x) t.erase(it);
}
int order_of_key(const T &x) const {
    return t.order_of_key({x, 0});
}
T find_by_order(int k) const {
    auto it = t.find_by_order(k);
    if (it == t.end()) return 1000000000;
    return it->first;
}
int size() const { return t.size(); }
bool empty() const { return t.empty(); }
};

```

SegmentTree.h

Description: Zero-indexed max-tree. Bounds are inclusive to the left and exclusive to the right. Can be changed by modifying T, f and unit.

Time: $\mathcal{O}(\log N)$ ea56ff, 36 lines

```

using T = ll;
struct Seg {
    int n, m=1;
    vector<T> t;
    const T unit = -inf;    // identity element

    Seg(int _n): n(_n) {
        while(m<n) m<=<=1;
        t.assign(2*m, unit);
    }
    T f(T a, T b) { return max(a, b); } // operation

    void upd(int p, T v) {
        for(t[p+=m]=v; p>1; p>>=1)
            t[p>>1] = f(t[p&~1], t[p|1]); // parent, lef, right
    }
    T qry(int l, int r) {    // [l, r)
        T L=unit, R=unit;
        for(l+=m, r+=m; l<r; l>>=1, r>>=1) {
            if(l&1) L = f(L, t[l++]);
            if(r&1) R = f(t[--r], R);
        }
        return f(L,R);
    }
    // first position with value >= v
    int first(T v) {
        if(t[1] < v) return -1;
        int p = 1;
        while(p < m) {
            if(t[2*p] >= v) p = 2*p;
            else p = 2*p+1;
        }
        int res = p - m;
        return (res < n) ? res : -1;
    }
};

```

LazySegmentTree.h

Description: can use set and add together. ranges are all [], EVEN IN CONSTRUCTOR ***

Usage: Lseg seg(0, n); Lseg seg(-50000, 50001)

Time: $\mathcal{O}(\log N)$. 2f3b1b, 61 lines

```

using T = ll;
struct Node {
    T mn=inf, mx=-inf, sm=0; // remove/add variables as needed
};
struct Lseg {
    Lseg *lc=0, *rc=0; Node v; int l, r;

```

```

// s:set tag (inf=none), a:add tag, v:stored node values
T s=inf, a=0; //custom

Node f(Node i, Node j) { // merge function // custom
    return {min(i.mn, j.mn), max(i.mx, j.mx), i.sm + j.sm};
}

// Huge interval of neutrals
Lseg(int _l, int _r): l(_l), r(_r) {}

void ps(T x) { // apply set // custom
    s=v.mn=v.mx=x, a=0, v.sm=x*(r-l);
}

void pa(T x) { // apply add //custom
    (s!=inf ? s : a)+=x, v.mn+=x, v.mx+=x, v.sm+=x*(r-l);
}

void ph() { // push
    int m = l+(r-l)/2;
    if(!lc) lc=new Lseg(l, m), rc=new Lseg(m, r);
    if(s!=inf) lc->ps(s), rc->ps(s), s=inf; //custom
    if(a) lc->pa(a), rc->pa(a), a=0; //custom
}

void set(int x, int y, T val) { // set [x, y) to val
    if(y<=l || r<=x) return;
    if(x<=l && r<=y) ps(val); //custom
    else ph(), lc->set(x, y, val), rc->set(x, y, val), \
        v=f(lc->v, rc->v);
}

void add(int x, int y, T val) { // Add val to [x, y)
    if(y<=l || r<=x) return;
    if(x<=l && r<=y) pa(val); //custom
    else ph(), lc->add(x, y, val), rc->add(x, y, val), \
        v=f(lc->v, rc->v);
}

Node qry(int x, int y) {
    if(y<=l || r<=x) return Node();
    if(x<=l && r<=y) return v;
    ph(); return f(lc->qry(x, y), rc->qry(x, y));
}

// First index in [x, y) with value >= val
int fst(T val, int x, int y) {
    if(y<=l || r<=x || v.mx<val) return -1;
    if(l+1 == r) return l;
    ph(); int res = lc->fst(val, x, y);
    return (res != -1) ? res : rc->fst(val, x, y);
}
~Lseg() { delete lc; delete rc; }
};

```

PersistentSeg.h

Description: ...

Time: ... 0d2541, 62 lines

```

static const ll INF = -(1LL << 60);
struct PersistentSegTree {
    struct Node {
        int l = 0, r = 0; ll mx = -INF; // change as needed
    };

    vector<Node> st;

    PersistentSegTree() {st.push_back(Node());} // 0 = null

```

```

int update(int prev, int tl, int tr, int pos, ll val) {
    st.push_back(st[prev]); // new copy of prev's root
    int cur = (int)st.size() - 1;
    if (tl == tr) {
        st[cur].mx = val;
        return cur;
    }
    int tm = (tl + tr) >> 1; // either l/r of root changes
    if (pos <= tm)
        st[cur].l = update(st[prev].l, tl, tm, pos, val);
    else
        st[cur].r = update(st[prev].r, tm + 1, tr, pos, val);

    st[cur].mx = max(st[st[cur].l].mx, st[st[cur].r].mx);
    return cur;
}

```

```

// query(root[i], 0, n-1, l, r) means
// maximum on range [l,r] in version i
ll query(int node, int tl, int tr, int l, int r) {
    if (!node || r < tl || tr < l) return -INF;
    if (l <= tl && tr <= r) return st[node].mx;

```

```

    int tm = (tl + tr) >> 1;
    return max(
        query(st[node].l, tl, tm, l, r),
        query(st[node].r, tm + 1, tr, l, r)
    );
}

```

// CUSTOM:

// Collect all ids in this version whose stored v >= need.
// Problem statement bounds total leaves accessed, so its not costly

```

void collect(int node, int tl, int tr, ll v_need, vector<
    int>& out) const {
    if (node == 0 || st[node].mx < v_need) return;
    if (tl == tr) {
        out.push_back(tl);
        return;
    }
    int tm = (tl + tr) >> 1;
    collect(st[node].l, tl, tm, v_need, out);
    collect(st[node].r, tm + 1, tr, v_need, out);
}
};

```

```

USAGE:
PersistentSegTree pst;
vector<int> root(N + 1, 0);
for (int i = N - 1; i >= 0; i--) { //or forward, whatever works
    root[i] = pst.update(root[i + 1], 0, N - 1, pos, val);
}

```

SegTree2D.h

Description: ...

Usage: one based grid, and all inclusive

Time: log square n 3e4e8d, 30 lines

```

const int TREE_SIZE = 1 << 10;
int trees[TREE_SIZE * 2][TREE_SIZE * 2];
// Adds d to the position x in the segment tree 'tree'.
void add(int *tree, int x, int d) {
    x += TREE_SIZE;
    while (x > 0) { tree[x] += d; x /= 2; }
}
// Adds d to the position (i, j).

```

```
void add(int y, int x, int d) {
    y += TREE_SIZE;
    while (y > 0) {    add(trees[y], x, d);    y /= 2;    }
}
// Returns the sum of value in range [x1, x2] in ‘tree’.
int query(int *tree, int x1, int x2) {
    x1 += TREE_SIZE;    x2 += TREE_SIZE;    int sum = 0;
    while (x1 <= x2) {
        if (x1 % 2 == 1) sum += tree[x1++];
        if (x2 % 2 == 0) sum += tree[x2--];
        x1 /= 2;    x2 /= 2;
    } return sum;
}
// Returns the sum of value in the rectangle [y1, y2]x[x1, x2]
int query(int y1, int y2, int x1, int x2) {
    y1 += TREE_SIZE;    y2 += TREE_SIZE;    int sum = 0;
    while (y1 <= y2) {
        if (y1 % 2 == 1) sum += query(trees[y1++], x1, x2);
        if (y2 % 2 == 0) sum += query(trees[y2--], x1, x2);
        y1 /= 2;    y2 /= 2;
    } return sum;
}
}
```

UnionFindRollback.h

Description: Disjoint-set data structure with undo. If undo is not needed, skip : st, time() and rollback().
Usage: int t = uf.time(); ...; uf.rollback(t);
Time: $\mathcal{O}(\log(N))$

05a554, 26 lines

```
// c = current no. of components
// time() returns current checkpoint
// rollback(x) rolls back to checkpoint x
struct DSU {
    vi e; vector<pii> st; int c;
    DSU(int n) : e(n, -1), c(n) {}
    int size(int x) { return -e[find(x)]; }
    int find(int x) { return e[x] < 0 ? x : find(e[x]); }
    int time() { return sz(st); }
    void rollback(int t) {
        c += (time() - t) / 2;
        for (int i = time(); i --> t; )
            e[st[i].first] = st[i].second;
        st.resize(t);
    }
    bool join(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) return false;
        if (e[a] > e[b]) swap(a, b);
        st.push_back({a, e[a]});
        st.push_back({b, e[b]});
        e[a] += e[b]; e[b] = a;
        --c;
        return true;
    }
};
```

SubMatrix.h

Description: Calculate submatrix sums quickly, given upper-left and lower-right corners (half-open).
Usage: SubMatrix<int> m(matrix);
m.sum(0, 0, 2, 2); // top left 4 elements
Time: $\mathcal{O}(N^2 + Q)$

c59ada, 13 lines

```
template<class T>
struct SubMatrix {
    vector<vector<T>> p;
    SubMatrix(vector<vector<T>>& v) {
        int R = sz(v), C = sz(v[0]);
        p.assign(R+1, vector<T>(C+1));
    }
};
```

```
rep(r,0,R) rep(c,0,C)
    p[r+1][c+1] = v[r][c] + p[r][c+1] + p[r+1][c] - p[r][c];
}
T sum(int u, int l, int d, int r) {
    return p[d][r] - p[d][l] - p[u][r] + p[u][l];
}
};
```

Matrix.h

Description: Basic operations on square matrices.

Usage: Matrix<int, 3> A;
A.d = {{{{1,2,3}}}, {{4,5,6}}, {{7,8,9}}}}};
array<int, 3> vec = {1,2,3};
vec = (A^N) * vec;

f71c23, 31 lines

```
template<class T, int N> struct Matrix {
    typedef Matrix M;
    array<array<T, N>, N> d{};
    M operator*(const M& m) const {
        M a;
        rep(i,0,N) rep(j,0,N)
            rep(k,0,N) a.d[i][k] += d[i][j] * m.d[j][k];
        return a;
    }
    array<T, N> operator*(const array<T, N>& vec) const {
        array<T, N> ret{};
        rep(i,0,N) rep(j,0,N) ret[i] += d[i][j] * vec[j];
        return ret;
    }
    static M identity() {
        M I;
        rep(i,0,N) I.d[i][i] = 1;
        return I;
    }
    M operator^(ll p) const {
        assert(p >= 0);
        M a, b(*this);
        rep(i,0,N) a.d[i][i] = 1;
        while (p) {
            if (p&1) a = a*b;
            b = b*b;
            p >>= 1;
        }
        return a;
    }
};
```

LineContainer.h

Description: Container where you can add lines of the form $kx+m$, and query maximum values at points x . Useful for dynamic programming (“convex hull trick”).
Time: $\mathcal{O}(\log N)$

Sec1c7, 30 lines

```
struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) { // floored division
        return a / b - ((a ^ b) < 0 && a % b); }
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
};
```

```
}
void add(ll k, ll m) {
    auto z = insert({k, m, 0}), y = z++, x = y;
    while (isect(y, z)) z = erase(z);
    if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
    while ((y = x) != begin() && (--x)->p >= y->p)
        isect(x, erase(y));
}
ll query(ll x) {
    assert(!empty());
    auto l = *lower_bound(x);
    return l.k * x + l.m;
}
};
```

Treap.h

Description: ...

Time: $\mathcal{O}(\log N)$

83ee02, 121 lines

```
// Indexing is 0-based
struct Treap {
    struct Node {
        int val, sum, size, prior, lazy;
        Node *l = nullptr, *r = nullptr;
        Node(int v):val(v),sum(v),size(1), prior(rand()), lazy(0){}
    };
};
```

Node* root = nullptr;

```
int _sz(Node* t) { return t ? t->size : 0; }
int sum(Node* t) { return t ? t->sum : 0; }
```

```
void push(Node* t) {
    if (!t || t->lazy == 0) return;
    t->val += t->lazy;
    t->sum += t->lazy * _sz(t);
    if (t->l) t->l->lazy += t->lazy;
    if (t->r) t->r->lazy += t->lazy;
    t->lazy = 0;
}
```

```
void upd(Node* t) {
    if (!t) return;
    push(t->l); push(t->r);
    t->size = 1 + _sz(t->l) + _sz(t->r);
    t->sum = t->val + sum(t->l) + sum(t->r);
}
```

```
void split(Node* t, int k, Node*& l, Node*& r) {
    // l holds first k indices [0...k-1]
    // r holds rem. indices [k...n-1]
```

```
    if (!t) { l = r = nullptr; return; }
    push(t);
    if (_sz(t->l) < k) {
        split(t->r, k - _sz(t->l) - 1, t->r, r);
        l = t;
    } else {
        split(t->l, k, l, t->l);
        r = t;
    }
    upd(t);
}
```

```
Node* merge(Node* l, Node* r) {
    push(l); push(r);
    if (!l || !r) return l ? l : r;
    if (l->prior > r->prior) {
        l->r = merge(l->r, r); upd(l); return l;
    }
}
```

```
    } else {
        r->l = merge(l, r->l); upd(r); return r;
    }
}

// inserts val at index pos [0 based], shifting the rest right
void insert(int pos, int val) {
    Node *t1, *t2;
    split(root, pos, t1, t2);
    root = merge(merge(t1, new Node(val)), t2);
}

// Range add [l, r]
void range_add(int l, int r, int v) {
    Node *t1, *t2, *t3; // [0...l-1], [l...r], [r+1...n-1]
    split(root, r+1, t2, t3);
    split(t2, l, t1, t2);
    if (t2) t2->lazy += v;
    root = merge(merge(t1, t2), t3);
}

// Range sum [l, r]
int range_sum(int l, int r) {
    Node *t1, *t2, *t3;
    split(root, r+1, t2, t3);
    split(t2, l, t1, t2);
    int ans = sum(t2);
    root = merge(merge(t1, t2), t3);
    return ans;
}

void erase(int pos) {
    // [0...pos-1], [pos...pos], [pos+1...n-1]
    Node *t1, *t2, *t3;
    split(root, pos + 1, t2, t3);
    split(t2, pos, t1, t2);
    delete t2;
    root = merge(t1, t3);
}

// add functions as needed :

void f(ll a, ll b) {
    if(a >= b) return;

    // both cases can be generalized to ...|block|...|block|...
    // any of the ... can be null, no problem

    Node *t1, *t2, *t3, *t4, *t5;
    int blocksize = min(n-b, b-a);
    split(root, b+blocksize, t4, t5);
    split(t4, b, t3, t4);
    split(t3, a+blocksize, t2, t3);
    split(t2, a, t1, t2);
    Node *res = merge(t1, t4);
    res = merge(res, t3);
    res = merge(res, t2);
    res = merge(res, t5);
    root = res;
}

void print() {
    print(root); cout << nl;
}

void print(Node* root) { // simply in-order
    if(!root) return;
    print(root->l);
    cout << root->val << gp;
    print(root->r);
}
```

```
    }
};

FenwickTree.h
Description: Computes partial sums a[0] + a[1] + ... + a[pos - 1], and
updates single elements a[i], taking the difference between the old and new
value.
Time: Both operations are  $\mathcal{O}(\log N)$ .
e62fac, 22 lines

struct FT {
    vector<ll> s;
    FT(int n) : s(n) {}
    void update(int pos, ll dif) { // a[pos] += dif
        for (; pos < sz(s); pos |= pos + 1) s[pos] += dif;
    }
    ll query(int pos) { // sum of values in [0, pos)
        ll res = 0;
        for (; pos > 0; pos &= pos - 1) res += s[pos-1];
        return res;
    }
    int lower_bound(ll sum) { // min pos st sum of [0, pos] >= sum
        // Returns n if no sum is >= sum, or -1 if empty sum is.
        if (sum <= 0) return -1;
        int pos = 0;
        for (int pw = 1 << 25; pw; pw >= 1) {
            if (pos + pw <= sz(s) && s[pos + pw-1] < sum)
                pos += pw, sum -= s[pos-1];
        }
        return pos;
    }
};

FenwickTree2d.h
Description: Computes sums a[i,j] for all i<I, j<J, and increases single elements a[i,j]. Requires that the elements to be updated are known in advance (call fakeUpdate() before init()).
Time:  $\mathcal{O}(\log^2 N)$ . (Use persistent segment trees for  $\mathcal{O}(\log N)$ .)
"FenwickTree.h" 157f07, 22 lines

struct FT2 {
    vector<vi> ys; vector<FT> ft;
    FT2(int limx) : ys(limx) {}
    void fakeUpdate(int x, int y) {
        for (; x < sz(ys); x |= x + 1) ys[x].push_back(y);
    }
    void init() {
        for (vi& v : ys) sort(all(v)), ft.emplace_back(sz(v));
    }
    int ind(int x, int y) {
        return (int)(lower_bound(all(ys[x]), y) - ys[x].begin()); }
    void update(int x, int y, ll dif) {
        for (; x < sz(ys); x |= x + 1)
            ft[x].update(ind(x, y), dif);
    }
    ll query(int x, int y) {
        ll sum = 0;
        for (; x; x &= x - 1)
            sum += ft[x-1].query(ind(x-1, y));
        return sum;
    }
};

RMQ.h
Description: Range Minimum Queries on an array. Returns min(V[a, V[a
+ 1], ... V[b - 1]) in constant time.
Usage: RMQ rmq(values);
rmq.query(inclusive, exclusive);
Time:  $\mathcal{O}(|V| \log |V| + Q)$ 
510c32, 16 lines

template<class T>
```

```
struct RMQ {
    vector<vector<T>> jmp;
    RMQ(const vector<T>& V) : jmp(1, V) {
        for (int pw = 1, k = 1; pw * 2 <= sz(V); pw *= 2, ++k) {
            jmp.emplace_back(sz(V) - pw * 2 + 1);
            rep(j, 0, sz(jmp[k]))
                jmp[k][j] = min(jmp[k - 1][j], jmp[k - 1][j + pw]);
        }
    }
    T query(int a, int b) {
        assert(a < b); // or return inf if a == b
        int dep = 31 - __builtin_clz(b - a);
        return min(jmp[dep][a], jmp[dep][b - (1 << dep)]);
    }
};

MoQueries.h
Description: Answer interval or tree path queries by finding an approximate TSP through the queries, and moving from one query to the next by adding/removing points at the ends. If values are on tree edges, change step to add/remove the edge (a,c) and remove the initial add call (but keep in).
Time:  $\mathcal{O}(N\sqrt{Q})$ 
a12ef4, 49 lines

void add(int ind, int end) { ... } // add a[ind] (end = 0 or 1)
void del(int ind, int end) { ... } // remove a[ind]
int calc() { ... } // compute current answer

vi mo(vector<pii> Q) {
    int L = 0, R = 0, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s;
    #define K(x) pii(x.first/blk, x.second ^ -(x.first/blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int s, int t){ return K(Q[s]) < K(Q[t]); });
    for (int qi : s) {
        pii q = Q[qi];
        while (L > q.first) add(--L, 0);
        while (R < q.second) add(R++, 1);
        while (L < q.first) del(L++, 0);
        while (R > q.second) del(--R, 1);
        res[qi] = calc();
    }
    return res;
}

vi moTree(vector<array<int, 2>> Q, vector<vi>& ed, int root=0){
    int N = sz(ed), pos[2] = {}, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s, I(N), L(N), R(N), in(N), par(N);
    add(0, 0), in[0] = 1;
    auto dfs = [&](int x, int p, int dep, auto& f) -> void {
        par[x] = p;
        L[x] = N;
        if (dep) I[x] = N++;
        for (int y : ed[x]) if (y != p) f(y, x, !dep, f);
        if (!dep) I[x] = N++;
        R[x] = N;
    };
    dfs(root, -1, 0, dfs);
    #define K(x) pii(I[x[0]] / blk, I[x[1]] ^ -(I[x[0]] / blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int s, int t){ return K(Q[s]) < K(Q[t]); });
    for (int qi : s) rep(end,0,2) {
        int &a = pos[end], b = Q[qi][end], i = 0;
        #define step(c) { if (in[c]) { del(a, end); in[a] = 0; } \
            else { add(c, end); in[c] = 1; } a = c; }
        while (! (L[b] <= L[a] && R[a] <= R[b]))
            I[i++] = b, b = par[b];
        while (a != b) step(par[a]);
        while (i--) step(I[i]);
        if (end) res[qi] = calc();
    }
}
```

```
    }
    return res;
}
```

WaveletTree.h

Description: Order statistics

Usage: ...

Time: ...

	7e165a, 83 lines
const int MAXV = (int)1e9 + 9;//max val of any element in array //array values can be negative too, so \ use appropriate minimum and maximum value	
struct wavelet_tree { int lo, hi; wavelet_tree *l, *r; int *b; // prefix sum of cnt <= mid long long *c; // prefix sum of elements int bsz, csz; wavelet_tree() : lo(1), hi(0), l(nullptr), r(nullptr),\ b(nullptr), c(nullptr), bsz(0), csz(0) {} void init(int *from, int *to, int x, int y) { lo = x; hi = y; if (from >= to) return ; int mid = (lo + hi) >> 1; auto f = [mid](int x){ return x <= mid; }; int n = to - from; b = (int*)malloc((n + 2) * sizeof(int)); c = (long long*)malloc((n + 2) * sizeof(long long)); b[0] = 0; c[0] = 0; for (int i = 0; i < n; i++) { b[i + 1] = b[i] + f(from[i]); c[i + 1] = c[i] + from[i]; // use long long } bsz = n + 1; csz = n + 1; if (lo == hi) return ; auto pivot = stable_partition(from, to, f); l = new wavelet_tree(); l->init(from, pivot, lo, mid); r = new wavelet_tree(); r->init(pivot, to, mid + 1, hi); } //kth smallest element in [l, r] int kth(int lpos, int rpos, int k) { if (lpos > rpos) return 0; if (lo == hi) return lo; int inLeft = b[rpos] - b[lpos - 1]; int lb = b[lpos - 1], rb = b[rpos]; if (k <= inLeft) return l->kth(lb + 1, rb, k); return r->kth(lpos - lb, rpos - rb, k - inLeft); } //count of numbers in [l, r] Less than or equal to k int LTE(int lpos, int rpos, int k) { if (lpos > rpos k < lo) return 0; if (hi <= k) return rpos - lpos + 1; int lb = b[lpos - 1], rb = b[rpos]; return l->LTE(lb + 1, rb, k) + \ r->LTE(lpos - lb, rpos - rb, k); } } //count of numbers in [l, r] equal to k int count(int lpos, int rpos, int k) { if (lpos > rpos k < lo k > hi) return 0; if (lo == hi) return rpos - lpos + 1; int lb = b[lpos - 1], rb = b[rpos]; int mid = (lo + hi) >> 1; if (k <= mid) return l->count(lb + 1, rb, k);	

```
        return r->count(lpos - lb, rpos - rb, k);  
    }  
    //sum of numbers in [l ,r] less than or equal to k  
    long long sum(int lpos, int rpos, int k) {  
        if (lpos > rpos || k < lo) return 0;  
        if (hi <= k) return c[rpos] - c[lpos - 1];  
        int lb = b[lpos - 1], rb = b[rpos];  
        return l->sum(lb + 1, rb, k) + \  
            r->sum(lpos - lb, rpos - rb, k);  
    }  
  
    ~wavelet_tree() {  
        if (l) delete l;  
        if (r) delete r;  
        if (b) free(b);  
        if (c) free(c);  
    }  
};  
// Usage:  
int a[n+1] = {0};  
L(i, 1, n) cin >> a[i];  
wavelet_tree wt;  
wt.init(a+1, a+n+1, 1, MAXV); //beware! after the init()  
operation array a[] will not be same
```

Numerical (3)

3.1 Polynomials and recurrences

Polynomial.h

	c9b7b0, 17 lines
struct Poly { vector<double> a; double operator() (double x) const { double val = 0; for (int i = sz(a); i--;) (val *= x) += a[i]; return val; } void diff() { rep(i,1,sz(a)) a[i-1] = i*a[i]; a.pop_back(); } void divroot(double x0) { double b = a.back(), c; a.back() = 0; for(int i=sz(a)-1; i--;) c = a[i], a[i] = a[i+1]*x0+b, b=c; a.pop_back(); } };	

PolyRoots.h

Description: Finds the real roots to a polynomial.

Usage: polyRoots({{2,-3,1}},-1e9,1e9) // solve x^2-3x+2 = 0

Time: $\mathcal{O}(n^2 \log(1/\epsilon))$

"Polynomial.h"	b00bfe, 23 lines
vector<double> polyRoots(Poly p, double xmin, double xmax) { if (sz(p.a) == 2) { return {-p.a[0]/p.a[1]}; } vector<double> ret; Poly der = p; der.diff(); auto dr = polyRoots(der, xmin, xmax); dr.push_back(xmin-1); dr.push_back(xmax+1); sort(all(dr)); rep(i,0,sz(dr)-1) { double l = dr[i], h = dr[i+1]; bool sign = p(l) > 0; if (sign ^ (p(h) > 0)) { rep(it,0,60) { // while (h - l > 1e-8)	

```
        double m = (l + h) / 2, f = p(m);  
        if ((f <= 0) ^ sign) l = m;  
        else h = m;  
    }  
    ret.push_back((l + h) / 2);  
}  
}  
return ret;  
}
```

PolyInterpolate.h

Description: Given n points $(x[i], y[i])$, computes an $n-1$ -degree polynomial p that passes through them: $p(x) = a[0] * x^0 + ... + a[n-1] * x^{n-1}$. For numerical precision, pick $x[k] = c * \cos(k/(n-1) * \pi)$, $k = 0 \dots n-1$.

Time: $\mathcal{O}(n^2)$

	08bf48, 13 lines
typedef vector<double> vd; vd interpolate(vd x, vd y, int n) { vd res(n), temp(n); rep(k,0,n-1) rep(i,k+1,n) y[i] = (y[i] - y[k]) / (x[i] - x[k]); double last = 0; temp[0] = 1; rep(k,0,n) rep(i,0,n) { res[i] += y[k] * temp[i]; swap(last, temp[i]); temp[i] -= last * x[k]; } return res; }	

BerlekampMassey.h

Description: Recovers any n -order linear recurrence relation from the first $2n$ terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. Should work on any field, but numerical stability for floats is not guaranteed. Output will have size $\leq n$.

Usage: berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}

Time: $\mathcal{O}(N^2)$

"../number-theory/ModPow.h"	96548b, 20 lines
vector<ll> berlekampMassey(vector<ll> s) { int n = sz(s), L = 0, m = 0; vector<ll> C(n), B(n), T; C[0] = B[0] = 1; ll b = 1; rep(i,0,n) { ++m; ll d = s[i] % mod; rep(j,1,L+1) d = (d + C[j] * s[i - j]) % mod; if (!d) continue ; T = C; ll coef = d * modpow(b, mod-2) % mod; rep(j,m,n) C[j] = (C[j] - coef * B[j - m]) % mod; if (2 * L > i) continue ; L = i + 1 - L; B = T; b = d; m = 0; } C.resize(L + 1); C.erase(C.begin()); for (ll& x : C) x = (mod - x) % mod; return C; }	

LinearRecurrence.h

Description: Generates the k 'th term of an n -order linear recurrence $S[i] = \sum_j S[i-j-1]tr[j]$, given $S[0 \dots \geq n-1]$ and $tr[0 \dots n-1]$. Faster than matrix multiplication. Useful together with Berlekamp-Massey.

Usage: linearRec({0, 1}, {1, 1}, k) // k 'th Fibonacci number

Time: $\mathcal{O}(n^2 \log k)$

	f4e444, 26 lines
typedef vector<ll> Poly; ll linearRec(Poly S, Poly tr, ll k) {	

```
int n = sz(tr);

auto combine = [&](Poly a, Poly b) {
    Poly res(n * 2 + 1);
    rep(i,0,n+1) rep(j,0,n+1)
        res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
    for (int i = 2 * n; i > n; --i) rep(j,0,n)
        res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
    res.resize(n + 1);
    return res;
};

Poly pol(n + 1), e(pol);
pol[0] = e[1] = 1;

for (++k; k; k /= 2) {
    if (k % 2) pol = combine(pol, e);
    e = combine(e, e);
}

ll res = 0;
rep(i,0,n) res = (res + pol[i + 1] * S[i]) % mod;
return res;
}
```

3.2 Matrices

Determinant.h

Description: Calculates determinant of a matrix. Destroys the matrix.
Time: $\mathcal{O}(N^3)$

```
double det(vector<vector<double>>& a) {
    int n = sz(a); double res = 1;
    rep(i,0,n) {
        int b = i;
        rep(j,i+1,n) if (fabs(a[j][i]) > fabs(a[b][i])) b = j;
        if (i != b) swap(a[i], a[b]), res *= -1;
        res *= a[i][i];
        if (res == 0) return 0;
        rep(j,i+1,n) {
            double v = a[j][i] / a[i][i];
            if (v != 0) rep(k,i+1,n) a[j][k] -= v * a[i][k];
        }
    }
    return res;
}
```

IntDeterminant.h

Description: Calculates determinant using modular arithmetics. Modulos can also be removed to get a pure-integer version.
Time: $\mathcal{O}(N^3)$

```
const ll mod = 12345;
ll det(vector<vector<ll>>& a) {
    int n = sz(a); ll ans = 1;
    rep(i,0,n) {
        rep(j,i+1,n) {
            while (a[j][i] != 0) { // gcd step
                ll t = a[i][i] / a[j][i];
                if (t) rep(k,i,n)
                    a[i][k] = (a[i][k] - a[j][k] * t) % mod;
                swap(a[i], a[j]);
                ans *= -1;
            }
        }
        ans = ans * a[i][i] % mod;
        if (!ans) return 0;
    }
    return (ans + mod) % mod;
}
```

SolveLinear.h

Description: Solves $A * x = b$. If there are multiple solutions, an arbitrary one is returned. Returns rank, or -1 if no solutions. Data in A and b is lost.
Time: $\mathcal{O}(n^2m)$

```
typedef vector<double> vd;
const double eps = 1e-12;

int solveLinear(vector<vd>& A, vd& b, vd& x) {
    int n = sz(A), m = sz(x), rank = 0, br, bc;
    if (n) assert(sz(A[0]) == m);
    vi col(m); iota(all(col), 0);

    rep(i,0,n) {
        double v, bv = 0;
        rep(r,i,n) rep(c,i,m)
            if ((v = fabs(A[r][c])) > bv)
                br = r, bc = c, bv = v;
        if (bv <= eps) {
            rep(j,i,n) if (fabs(b[j]) > eps) return -1;
            break;
        }
        swap(A[i], A[br]);
        swap(b[i], b[br]);
        swap(col[i], col[bc]);
        rep(j,0,n) swap(A[j][i], A[j][bc]);
        bv = 1/A[i][i];
        rep(j,i+1,n) {
            double fac = A[j][i] * bv;
            b[j] -= fac * b[i];
            rep(k,i+1,m) A[j][k] -= fac*A[i][k];
        }
        rank++;
    }

    x.assign(m, 0);
    for (int i = rank; i--;) {
        b[i] /= A[i][i];
        x[col[i]] = b[i];
        rep(j,0,i) b[j] -= A[j][i] * b[i];
    }
    return rank; // (multiple solutions if rank < m)
}
```

SolveLinear2.h

Description: To get all uniquely determined values of x back from SolveLinear, make the following changes:

```
"SolveLinear.h"
08e495, 7 lines

rep(j,0,n) if (j != i) // instead of rep(j,i+1,n)
// ... then at the end:
x.assign(m, undefined);
rep(i,0,rank) {
    rep(j,rank,m) if (fabs(A[i][j]) > eps) goto fail;
    x[col[i]] = b[i] / A[i][i];
fail:; }
```

SolveLinearBinary.h

Description: Solves $Ax = b$ over $\mathbb{F}_2$. If there are multiple solutions, one is returned arbitrarily. Returns rank, or -1 if no solutions. Destroys A and b .
Time: $\mathcal{O}(n^2m)$

```
typedef bitset<1000> bs;

int solveLinear(vector<bs>& A, vi& b, bs& x, int m) {
    int n = sz(A), rank = 0, br;
    assert(m <= sz(x));
    vi col(m); iota(all(col), 0);
    rep(i,0,n) {
        for (br=i; br<n; ++br) if (A[br].any()) break;
```

```
if (br == n) {
    rep(j,i,n) if(b[j]) return -1;
    break;
}
int bc = (int)A[br]._Find_next(i-1);
swap(A[i], A[br]);
swap(b[i], b[br]);
swap(col[i], col[bc]);
rep(j,0,n) if (A[j][i] != A[j][bc]) {
    A[j].flip(i); A[j].flip(bc);
}
rep(j,i+1,n) if (A[j][i]) {
    b[j] ^= b[i];
    A[j] ^= A[i];
}
rank++;
}

x = bs();
for (int i = rank; i--;) {
    if (!b[i]) continue;
    x[col[i]] = 1;
    rep(j,0,i) b[j] ^= A[j][i];
}
return rank; // (multiple solutions if rank < m)
}
```

MatrixInverse.h

Description: Invert matrix A . Returns rank; result is stored in A unless singular (rank < n). Can easily be extended to prime moduli; for prime powers, repeatedly set $A^{-1} = A^{-1}(2I - AA^{-1}) \pmod{p^k}$ where A^{-1} starts as the inverse of $A \bmod p$, and k is doubled in each step.
Time: $\mathcal{O}(n^3)$

```
ebfff6, 35 lines

int matInv(vector<vector<double>>& A) {
    int n = sz(A); vi col(n);
    vector<vector<double>> tmp(n, vector<double>(n));
    rep(i,0,n) tmp[i][i] = 1, col[i] = i;

    rep(i,0,n) {
        int r = i, c = i;
        rep(j,i,n) rep(k,i,n)
            if (fabs(A[j][k]) > fabs(A[r][c]))
                r = j, c = k;
        if (fabs(A[r][c]) < 1e-12) return i;
        A[i].swap(A[r]); tmp[i].swap(tmp[r]);
        rep(j,0,n)
            swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c]);
        swap(col[i], col[c]);
        double v = A[i][i];
        rep(j,i+1,n) {
            double f = A[j][i] / v;
            A[j][i] = 0;
            rep(k,i+1,n) A[j][k] -= f*A[i][k];
            rep(k,0,n) tmp[j][k] -= f*tmp[i][k];
        }
        rep(j,i+1,n) A[i][j] /= v;
        rep(j,0,n) tmp[i][j] /= v;
        A[i][i] = 1;
    }

    for (int i = n-1; i > 0; --i) rep(j,0,i) {
        double v = A[j][i];
        rep(k,0,n) tmp[j][k] -= v*tmp[i][k];
    }

    rep(i,0,n) rep(j,0,n) A[col[i]][col[j]] = tmp[i][j];
    return n;
}
```

Tridiagonal.h

Description: $x = \text{tridiagonal}(d, p, q, b)$ solves the equation system

$$\begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_{n-1} \end{pmatrix} = \begin{pmatrix} d_0 & p_0 & 0 & 0 & \cdots & 0 \\ q_0 & d_1 & p_1 & 0 & \cdots & 0 \\ 0 & q_1 & d_2 & p_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & q_{n-3} & d_{n-2} & p_{n-2} \\ 0 & 0 & \cdots & 0 & q_{n-2} & d_{n-1} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \end{pmatrix}.$$

This is useful for solving problems on the type

$$a_i = b_i a_{i-1} + c_i a_{i+1} + d_i, \, 1 \leq i \leq n,$$

where a_0, a_{n+1}, b_i, c_i and d_i are known. a can then be obtained from

$$\{a_i\} = \text{tridiagonal}(\{1, -1, -1, \dots, -1, 1\}, \{0, c_1, c_2, \dots, c_n\}, \{b_1, b_2, \dots, b_n, 0\}, \{a_0, d_1, d_2, \dots, d_n, a_{n+1}\}).$$

Fails if the solution is not unique.

If $|d_i| > |p_i| + |q_{i-1}|$ for all i , or $|d_i| > |p_{i-1}| + |q_i|$, or the matrix is positive definite, the algorithm is numerically stable and neither `tr` nor the check for `diag[i] == 0` is needed.

Time: $\mathcal{O}(N)$

```
typedef double T;
vector<T> tridiagonal(vector<T> diag, const vector<T>& super,
    const vector<T>& sub, vector<T> b) {
    int n = sz(b); vi tr(n);
    rep(i,0,n-1) {
        if (abs(diag[i]) < 1e-9 * abs(super[i])) { // diag[i] == 0
            b[i+1] -= b[i] * diag[i+1] / super[i];
            if (i+2 < n) b[i+2] -= b[i] * sub[i+1] / super[i];
            diag[i+1] = sub[i]; tr[++i] = 1;
        } else {
            diag[i+1] -= super[i]*sub[i]/diag[i];
            b[i+1] -= b[i]*sub[i]/diag[i];
        }
    }
    for (int i = n; i--;) {
        if (tr[i]) {
            swap(b[i], b[i-1]);
            diag[i-1] = diag[i];
            b[i] /= super[i-1];
        } else {
            b[i] /= diag[i];
            if (i) b[i-1] -= b[i]*super[i-1];
        }
    }
    return b;
}
```

8f9fa8, 26 lines

3.3 Fourier transforms

FastFourierTransform.h

Description: `fft(a)` computes $\hat{f}(k) = \sum_x a[x] \exp(2\pi i \cdot kx/N)$ for all k . N must be a power of 2. Useful for convolution: `conv(a, b) = c`, where $c[x] = \sum a[i]b[x-i]$. For convolution of complex numbers or more than two vectors: FFT, multiply pointwise, divide by `n`, `reverse(start+1, end)`, FFT back. Rounding is safe if $(\sum a_i^2 + \sum b_i^2) \log_2 N < 9 \cdot 10^{14}$ (in practice 10^{16} ; higher for random inputs). Otherwise, use NTT/FFTMod.

Time: $\mathcal{O}(N \log N)$ with $N = |A| + |B|$ ($\sim 1s$ for $N = 2^{22}$)

00ced6, 35 lines

```
typedef complex<double> C;
typedef vector<double> vd;
void fft(vector<C>& a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1);
    static vector<C> rt(2, 1); // (^ 10% faster if double)
    for (static int k = 2; k < n; k *= 2) {
        R.resize(n); rt.resize(n);
        auto x = polar(1.0L, acos(-1.0L) / k);
```

```
        rep(i,k,2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
    }
    vi rev(n);
    rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
            C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-rolled)
            a[i + j + k] = a[i + j] - z;
            a[i + j] += z;
        }
    }
    vd conv(const vd& a, const vd& b) {
        if (a.empty() || b.empty()) return {};
        vd res(sz(a) + sz(b) - 1);
        int L = 32 - __builtin_clz(sz(res)), n = 1 << L;
        vector<C> in(n), out(n);
        copy(all(a), begin(in));
        rep(i,0,sz(b)) in[i].imag(b[i]);
        fft(in);
        for (C& x : in) x *= x;
        rep(i,0,n) out[i] = in[-i & (n - 1)] - conj(in[i]);
        fft(out);
        rep(i,0,sz(res)) res[i] = imag(out[i]) / (4 * n);
        return res;
    }
```

FastFourierTransformMod.h

Description: Higher precision FFT, can be used for convolutions modulo arbitrary integers as long as $N \log_2 N \cdot \text{mod} < 8.6 \cdot 10^{14}$ (in practice 10^{16} or higher). Inputs must be in $[0, \text{mod})$.

Time: $\mathcal{O}(N \log N)$, where $N = |A| + |B|$ (twice as slow as NTT or FFT)

"FastFourierTransform.h" b82773, 22 lines

```
typedef vector<ll> vl;
template<int M> vl convMod(const vl &a, const vl &b) {
    if (a.empty() || b.empty()) return {};
    vl res(sz(a) + sz(b) - 1);
    int B=32-__builtin_clz(sz(res)), n=1<<B, cut=int(sqrt(M));
    vector<C> L(n), R(n), outs(n), outl(n);
    rep(i,0,sz(a)) L[i] = C((int)a[i] / cut, (int)a[i] % cut);
    rep(i,0,sz(b)) R[i] = C((int)b[i] / cut, (int)b[i] % cut);
    fft(L), fft(R);
    rep(i,0,n) {
        int j = -i & (n - 1);
        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
    }
    fft(outl), fft(outs);
    rep(i,0,sz(res)) {
        ll av = ll(real(outl[i])+.5), cv = ll(imag(outs[i])+.5);
        ll bv = ll(imag(outl[i])+.5) + ll(real(outs[i])+.5);
        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    }
    return res;
}
```

NumberTheoreticTransform.h

Description: `ntt(a)` computes $\hat{f}(k) = \sum_x a[x] g^{xk}$ for all k , where $g = \text{root}^{(\text{mod}-1)/N}$. N must be a power of 2. Useful for convolution modulo specific nice primes of the form $2^a b + 1$, where the convolution result has size at most 2^a . For arbitrary modulo, see FFTMod. `conv(a, b) = c`, where $c[x] = \sum a[i]b[x-i]$. For manual convolution: NTT the inputs, multiply pointwise, divide by `n`, `reverse(start+1, end)`, NTT back. Inputs must be in $[0, \text{mod})$.

Time: $\mathcal{O}(N \log N)$

"../number-theory/ModPow.h" ced03d, 34 lines

```
const ll mod = (119 << 23) + 1, root = 62; // = 998244353
```

```
typedef vector<ll> vl;
void ntt(vl &a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vl rt(2, 1);
    for (static int k = 2, s = 2; k < n; k *= 2, s++) {
        rt.resize(n);
        ll z[] = {1, modpow(root, mod >> s)};
        rep(i,k,2*k) rt[i] = rt[i / 2] * z[i & 1] % mod;
    }
    vi rev(n);
    rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
            ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
            a[i + j + k] = ai - z + (z > ai ? mod : 0);
            ai += (ai + z >= mod ? z - mod : z);
        }
    }
    vl conv(const vl &a, const vl &b) {
        if (a.empty() || b.empty()) return {};
        int s = sz(a) + sz(b) - 1, B = 32 - __builtin_clz(s),
            n = 1 << B;
        int inv = modpow(n, mod - 2);
        vl L(a), R(b), out(n);
        L.resize(n), R.resize(n);
        ntt(L), ntt(R);
        rep(i,0,n)
            out[-i & (n - 1)] = (ll)L[i] * R[i] % mod * inv % mod;
        ntt(out);
        return {out.begin(), out.begin() + s};
    }
```

FastSubsetTransform.h

Description: Transform to a basis with fast convolutions of the form $c[z] = \sum_{z=x \oplus y} a[x] \cdot b[y]$, where $\oplus$ is one of AND, OR, XOR. The size of a must be a power of two.

Time: $\mathcal{O}(N \log N)$

464cf3, 16 lines

```
void FST(vi& a, bool inv) {
    for (int n = sz(a), step = 1; step < n; step *= 2) {
        for (int i = 0; i < n; i += 2 * step) rep(j,i,i+step) {
            int &u = a[j], &v = a[j + step]; tie(u, v) =
                inv ? pii(v - u, u) : pii(v, u + v); // AND
                inv ? pii(v, u - v) : pii(u + v, u); // OR
                pii(u + v, u - v); // XOR
        }
    }
    if (inv) for (int& x : a) x /= sz(a); // XOR only
}
vi conv(vi a, vi b) {
    FST(a, 0); FST(b, 0);
    rep(i,0,sz(a)) a[i] *= b[i];
    FST(a, 1); return a;
}
```

Number theory (4)

4.1 Modular arithmetic

ModPow.h

b83e45, 8 lines

```
const ll mod = 1000000007; // faster if const
```

```
ll modpow(ll b, ll e) {
    ll ans = 1;
    for (; e; b = b * b % mod, e /= 2)
        if (e & 1) ans = ans * b % mod;
```

```
    return ans;
}
```

ModLog.h

Description: Returns the smallest $x > 0$ s.t. $a^x = b \pmod m$, or -1 if no such x exists. modLog(a,1,m) can be used to calculate the order of a .

Time: $\mathcal{O}(\sqrt{m})$ c040b8, 11 lines

```
ll modLog(ll a, ll b, ll m) {
    ll n = (ll) sqrt(m) + 1, e = 1, f = 1, j = 1;
    unordered_map<ll, ll> A;
    while (j <= n && (e = f * e * a % m) != b % m)
        A[e * b % m] = j++;
    if (e == b % m) return j;
    if (__gcd(m, e) == __gcd(m, b))
        rep(i,2,n+2) if (A.count(e = e * f % m))
            return n * i - A[e];
    return -1;
}
```

ModSum.h

Description: Sums of mod'ed arithmetic progressions.
 $\text{modsum}(\text{to}, c, k, m) = \sum_{i=0}^{\text{to}-1} (ki+c)\%m$. divsum is similar but for floored division.

Time: $\log(m)$, with a large constant. 5c5bc5, 16 lines

```
typedef unsigned long long ull;
ull sumsq(ull to) { return to / 2 * ((to-1) | 1); }
```

```
ull divsum(ull to, ull c, ull k, ull m) {
    ull res = k / m * sumsq(to) + c / m * to;
    k %= m; c %= m;
    if (!k) return res;
    ull to2 = (to * k + c) / m;
    return res + (to - 1) * to2 - divsum(to2, m-1 - c, m, k);
}
```

```
ll modsum(ull to, ll c, ll k, ll m) {
    c = ((c % m) + m) % m;
    k = ((k % m) + m) % m;
    return to * c + k * sumsq(to) - m * divsum(to, c, k, m);
}
```

ModMulLL.h

Description: Calculate $a \cdot b \bmod c$ (or $a^b \bmod c$) for $0 \leq a, b \leq c \leq 7.2 \cdot 10^{18}$.

Time: $\mathcal{O}(1)$ for modmul, $\mathcal{O}(\log b)$ for modpow bbbd8f, 11 lines

```
typedef unsigned long long ull;
ull modmul(ull a, ull b, ull M) {
    ll ret = a * b - M * ull(1.L / M * a * b);
    return ret + M * (ret < 0) - M * (ret >= (ll)M);
}
ull modpow(ull b, ull e, ull mod) {
    ull ans = 1;
    for (; e; b = modmul(b, b, mod), e /= 2)
        if (e & 1) ans = modmul(ans, b, mod);
    return ans;
}
```

ModSqrt.h

Description: Tonelli-Shanks algorithm for modular square roots. Finds x s.t. $x^2 = a \pmod p$ ($-x$ gives the other solution).

Time: $\mathcal{O}(\log^2 p)$ worst case, $\mathcal{O}(\log p)$ for most p

"ModPow.h" 19a793, 24 lines

```
ll sqrt(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    assert(modpow(a, (p-1)/2, p) == 1); // else no solution
    if (p % 4 == 3) return modpow(a, (p+1)/4, p);
```

```
// a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
ll s = p - 1, n = 2;
int r = 0, m;
while (s % 2 == 0)
    ++r, s /= 2;
while (modpow(n, (p - 1) / 2, p) != p - 1) ++n;
ll x = modpow(a, (s + 1) / 2, p);
ll b = modpow(a, s, p), g = modpow(n, s, p);
for (; r = m) {
    ll t = b;
    for (m = 0; m < r && t != 1; ++m)
        t = t * t % p;
    if (m == 0) return x;
    ll gs = modpow(g, 1LL << (r - m - 1), p);
    g = gs * gs % p;
    x = x * gs % p;
    b = b * g % p;
}
}
```

4.2 Primality

FastEratosthenes.h

Description: Prime sieve for generating all primes smaller than LIM.

Time: LIM=1e9 $\approx$ 1.5s 6b2912, 20 lines

```
const int LIM = 1e6;
bitset<LIM> isPrime;
vi eratosthenes() {
    const int S = (int)round(sqrt(LIM)), R = LIM / 2;
    vi pr = {2}, sieve(S+1); pr.reserve((int)(LIM/log(LIM)*1.1));
    vector<pii> cp;
    for (int i = 3; i <= S; i += 2) if (!sieve[i]) {
        cp.push_back({i, i * i / 2});
        for (int j = i * i; j <= S; j += 2 * i) sieve[j] = 1;
    }
    for (int L = 1; L <= R; L += S) {
        array<bool, S> block{};
        for (auto &[p, idx] : cp)
            for (int i=idx; i < S+L; idx = (i+=p)) block[i-L] = 1;
        rep(i,0,min(S, R - L))
            if (!block[i]) pr.push_back((L + i) * 2 + 1);
    }
    for (int i : pr) isPrime[i] = 1;
    return pr;
}
```

MillerRabin.h

Description: Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to $7 \cdot 10^{18}$; for larger numbers, use Python and extend A randomly.

Time: 7 times the complexity of $a^b \bmod c$.

"ModMulLL.h" 60dcd1, 12 lines

```
bool isPrime(ull n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    ull A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022},
        s = __builtin_ctzll(n-1), d = n >> s;
    for (ull a : A) { // ^ count trailing zeroes
        ull p = modpow(a%n, d, n), i = s;
        while (p != 1 && p != n - 1 && a % n && i--)
            p = modmul(p, p, n);
        if (p != n-1 && i != s) return 0;
    }
    return 1;
}
```

Factor.h

Description: Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).

Time: $\mathcal{O}(n^{1/4})$, less for numbers with small factors.

"ModMulLL.h", "MillerRabin.h" d8d98d, 18 lines

```
ull pollard(ull n) {
    ull x = 0, y = 0, t = 30, prd = 2, i = 1, q;
    auto f = [&](ull x) { return modmul(x, x, n) + i; };
    while (t++ % 40 || __gcd(prd, n) == 1) {
        if (x == y) x = ++i, y = f(x);
        if ((q = modmul(prd, max(x,y) - min(x,y), n))) prd = q;
        x = f(x), y = f(f(y));
    }
    return __gcd(prd, n);
}
vector<ull> factor(ull n) {
    if (n == 1) return {};
    if (isPrime(n)) return {n};
    ull x = pollard(n);
    auto l = factor(x), r = factor(n / x);
    l.insert(l.end(), all(r));
    return l;
}
```

spf.h

Description: ...

Time: ... bcd355, 16 lines

```
const int N = 1e7;
vector<int> spf(N);
void prep(){ // modified sieve
    L(i, 2, N-1) {
        if(spf[i] == 0) {
            spf[i] = i;
            for(ll j = i * i; j <= N-1; j += i) {
                if(spf[j] == 0) spf[j] = i;
            }}
    }
    // prime factorize [if you need 1e9, use loop upto sqrt]
    vector<pll> prime_factorize(ll x) {
        vector<pll> ans;
        while(x!=1) {
            ll cur = spf[x]; ll cnt = 0;
            while(x%cur==0) cnt++, x/=cur;
            ans.push_back({cur, cnt});
        }
        return ans;
    }
```

LinearSieve.h

Description: ...

Time: ... 84f2ca, 34 lines

```
const int N = 1000000;
vector<int> primes;
int spf[N + 1]; int phi[N + 1]; int mu[N + 1];
bool is_composite[N + 1];
// int func[MAXN], cnt[MAXN]; // Extension
```

```
void linear_sieve(int n) {
    std::fill(is_composite, is_composite + n, false);
    phi[1] = 1; mu[1] = 1;
    for (int i = 2; i <= n; i++) {
        if (!is_composite[i]) {
            primes.push_back(i);
            spf[i] = i; phi[i] = i - 1; mu[i] = -1;
        }
        for (int p : primes) {
            if (i * 1LL * p > n) break;
            is_composite[i * p] = true;
            spf[i * p] = p;

            if (i % p == 0) {
                phi[i * p] = phi[i] * p;
                mu[i * p] = 0;
                // func[i*prime[j]] = func[i]/cnt[i] * (cnt[i]+1);
            }
        }
    }
}
```

```
// cnt[i * prime[j]] = cnt[i] + 1;
    break;
} else {
    phi[i * p] = phi[i] * (p - 1);
    mu[i * p] = -mu[i];
// func[i * prime[j]] = func[i] * func[prime[j]];
// cnt[i * prime[j]] = 1;
}
}
}
```

4.3 Divisibility

euclid.h
Description: Finds two integers x and y , such that $ax + by = \gcd(a, b)$. If you just need gcd, use the built in `_gcd` instead. If a and b are coprime, then x is the inverse of $a \pmod b$.

```
// Extended Euclid
ll euclid(ll a, ll b, ll &x, ll &y) {
    if (b == 0) { x = 1; y = 0; return a; }
    ll x1, y1, g = euclid(b, a % b, x1, y1);
    x = y1; y = x1 - (a / b) * y1;
    return g;
}
```

Diophantine.h
Description: ...

```
const ll INF = 2e18;

// Solve ax + by = c
bool diophantine(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
    if (a == 0 && b == 0) {
        if (c == 0) { x = 0; y = 0; g = 0; return true; }
        return false;
    }
    g = euclid(abs(a), abs(b), x, y);
    if (c % g != 0) return false;

    if (a < 0) x = -x;
    if (b < 0) y = -y;

    x *= (c / g);
    y *= (c / g);

    return true;
}
```

DiophNonNeg.h
Description: ...

```
ll floor_div(ll a, ll b) {
    ll res = a / b;
    ll rem = a % b;
    if (rem != 0 && ((a < 0) ^ (b < 0))) res--;
    return res;
}
ll ceil_div(ll a, ll b) {
    ll res = a / b;
    ll rem = a % b;
    if (rem != 0 && !((a < 0) ^ (b < 0))) res++;
    return res;
}
// Non-negative solution (x >= 0, y >= 0)
bool dioph_nonneg(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
    if (!diophantine(a, b, c, x, y, g)) return false;

    if (a == 0) {
```

euclid Diophantine DiophNonNeg CRT phiFunction

```
    if (b == 0) return c == 0;
    if (y >= 0) { x = 0; return true; }
    return false;
}

if (b == 0) {
    if (x >= 0) { y = 0; return true; }
    return false;
}

ll dx = b / g;
ll dy = a / g;

ll k_min = -INF;
ll k_max = INF;

// x + k*dx >= 0 => k*dx >= -x
if (dx > 0) {
    k_min = max(k_min, ceil_div(-x, dx));
} else if (dx < 0) {
    k_max = min(k_max, floor_div(-x, dx));
} else if (-x > 0) {
    return false;
}

// y - k*dy >= 0 => k*dy <= y
if (dy > 0) {
    k_max = min(k_max, floor_div(y, dy));
} else if (dy < 0) {
    k_min = max(k_min, ceil_div(y, dy));
} else if (y < 0) {
    return false;
}

if (k_min > k_max) return false;

// Any k in [k_min, k_max] gives a non-negative solution.
// k_min minimizes x (if dx > 0) or maximizes x (if dx < 0)
ll k = k_min;
x += k * dx;
y -= k * dy;

return true;
}
```

```
CRT.h
Description: Chinese Remainder Theorem.
crt(a, m, b, n) computes x such that x ≡ a (mod m), x ≡ b (mod n). If
|a| < m and |b| < n, x will obey 0 ≤ x < lcm(m, n). Assumes mn < 2^62.
Time: log(n)
"euclid.h"
04d93a, 7 lines

ll crt(ll a, ll m, ll b, ll n) {
    if (n > m) swap(a, b), swap(m, n);
    ll x, y, g = euclid(m, n, x, y);
    assert((a - b) % g == 0); // else no solution
    x = (b - a) % n * x % n / g * m + a;
    return x < 0 ? x + m*n/g : x;
}
```

4.3.1 Bézout’s identity

For $a \neq 0, b \neq 0$, then $d = \gcd(a, b)$ is the smallest positive integer for which there are integer solutions to

$$ax + by = d$$

If (x, y) is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right), \quad k \in \mathbb{Z}$$

phiFunction.h
Description: Euler’s ϕ function is defined as $\phi(n) := \#$ of positive integers $\leq n$ that are coprime with n . $\phi(1) = 1$, p prime $\Rightarrow \phi(p^k) = (p - 1)p^{k-1}$, m, n coprime $\Rightarrow \phi(mn) = \phi(m)\phi(n)$. If $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$ then $\phi(n) = (p_1 - 1)p_1^{k_1-1} \dots (p_r - 1)p_r^{k_r-1}$. $\phi(n) = n \cdot \prod_{p|n} (1 - 1/p)$. $\sum_{d|n} \phi(d) = n$, $\sum_{1 \leq k \leq n, \gcd(k,n)=1} k = n\phi(n)/2, n > 1$
Euler’s thm: a, n coprime $\Rightarrow a^{\phi(n)} \equiv 1 \pmod n$.
Fermat’s little thm: p prime $\Rightarrow a^{p-1} \equiv 1 \pmod p \forall a$.

```
const int LIM = 5000000;
int phi[LIM];

void calculatePhi() {
    rep(i, 0, LIM) phi[i] = i&1 ? i : i/2;
    for (int i = 3; i < LIM; i += 2) if (phi[i] == i)
        for (int j = i; j < LIM; j += i) phi[j] -= phi[j] / i;
}
```

4.4 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$$

with $m > n > 0, k > 0, m \perp n$, and either m or n even.

4.5 Primes

$p = 962592769$ is such that $2^{21} \mid p - 1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a , except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

4.6 Estimates

$$\sum_{d|n} d = O(n \log \log n).$$

The number of divisors of n is at most around 100 for $n < 5e4$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

4.7 Mobius Function

$$\mu(n) = \begin{cases} 0 & n \text{ is not square free} \\ 1 & n \text{ has even number of prime factors} \\ -1 & n \text{ has odd number of prime factors} \end{cases}$$

Mobius Inversion:

$$g(n) = \sum_{d|n} f(d) \Leftrightarrow f(n) = \sum_{d|n} \mu(d)g(n/d)$$

Other useful formulas/forms:

$$\sum_{d|n} \mu(d) = [n = 1] \text{ (very useful)}$$

$$g(n) = \sum_{n|d} f(d) \Leftrightarrow f(n) = \sum_{n|d} \mu(d/n)g(d)$$

$$g(n) = \sum_{1 \leq m \leq n} f(\lfloor \frac{n}{m} \rfloor) \Leftrightarrow f(n) = \sum_{1 \leq m \leq n} \mu(m)g(\lfloor \frac{n}{m} \rfloor)$$

Combinatorial (5)

5.1 Permutations

5.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800
n	11	12	13				14	15	16	17
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14			
n	20	25	30	40	50	100	150	171		
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

IntPerm.h
Description: Permutation -> integer conversion. (Not order preserving.)
Integer -> permutation can use a lookup table.
Time: $\mathcal{O}(n)$

```
int permToInt(vi& v) {
    int use = 0, i = 0, r = 0;
    for(int x:v) r = r * ++i + __builtin_popcount(use & -(1<<x)),
        use |= 1 << x;
    return r;
}
```

5.1.2 Cycles

Let $g_S(n)$ be the number of n -permutations whose cycle lengths all belong to the set S . Then

$$\sum_{n=0}^\infty g_S(n) \frac{x^n}{n!} = \exp\left(\sum_{n \in S} \frac{x^n}{n}\right)$$

5.1.3 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1) + D(n-2)) = nD(n-1) + (-1)^n = \left\lfloor \frac{n!}{e} \right\rfloor$$

5.1.4 Burnside’s lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k|n} f(k) \phi(n/k).$$

5.2 Partitions and subsets

5.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

<i>n</i>	0	1	2	3	4	5	6	7	8	9	20	50	100
<i>p</i> (<i>n</i>)	1	1	2	3	5	7	11	15	22	30	627	~2e5	~2e8

5.2.2 Lucas’ Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + \dots + n_1 p + n_0$ and $m = m_k p^k + \dots + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod{p}$.

5.2.3 Hockey Stick Theorem

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$

5.2.4 Vandermonde’s Theorem

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

5.2.5 Binomial Coefficient Identity

$$(x+1) \binom{x}{k-1} = k \binom{x+1}{k}$$

5.2.6 Binomials

multinomial.h
Description: Computes $\binom{k_1 + \dots + k_n}{k_1, k_2, \dots, k_n} = \frac{(\sum k_i)!}{k_1! k_2! \dots k_n!}$.

```
ll multinomial(vi& v) {
    ll c = 1, m = v.empty() ? 1 : v[0];
    rep(i, 1, sz(v)) rep(j, 0, v[i]) c = c * ++m / (j+1);
    return c;
}
```

5.3 General purpose numbers

5.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able).
 $B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$

Sums of powers:

$$\sum_{i=1}^n n^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^\infty f(i) = \int_m^\infty f(x) dx - \sum_{k=1}^\infty \frac{B_k}{k!} f^{(k-1)}(m) \\ \approx \int_m^\infty f(x) dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

5.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k), \quad c(0, 0) = 1 \\ \sum_{k=0}^n c(n, k) x^k = x(x+1) \dots (x+n-1)$$

$c(8, k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$
 $c(n, 2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots$

5.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j :s s.t. $\pi(j) > \pi(j+1)$, $k+1$ j :s s.t. $\pi(j) \geq j$, k j :s s.t. $\pi(j) > j$.

$$E(n, k) = (n-k)E(n-1, k-1) + (k+1)E(n-1, k)$$

$$E(n, 0) = E(n, n-1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

5.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n, k) = S(n-1, k-1) + kS(n-1, k)$$

$$S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

5.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$. For p prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod{p}$$

5.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i : $n_1 n_2 \dots n_k n^{k-2}$
with degrees d_i : $(n-2)! / ((d_1-1)! \dots (d_n-1)!)$

5.3.7 Catalan numbers

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$
$$C_0 = 1, \ C_{n+1} = \frac{2(2n+1)}{n+2} C_n, \ C_{n+1} = \sum C_i C_{n-i}$$

$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \dots$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n + 1$ leaves (0 or 2 children).
- ordered trees with $n + 1$ vertices.
- ways a convex polygon with $n + 2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Graph (6)

6.1 Fundamentals

Dijkstra.h

Description: ...

Time: ...

bbab71, 15 lines

```
vector<ll> dijkstra(ll start, vector<vector<array<ll, 2>>> &gr,
    int n){
    vector<ll> d(n); For(i, 1, n) d[i] = inf; d[start] = 0;
    bool vis[n+1] = {0};
    priority_queue<array<ll, 2>, vector<array<ll, 2>>, greater<
        array<ll, 2>>> pq;
    pq.push({0LL, start});
    while(!pq.empty()){
        auto [dist, node] = pq.top(); pq.pop();
        if(vis[node]) continue;
        vis[node] = 1;
        for(auto [nextNode, weight] : gr[node]){
            if(weight + dist < d[nextNode]){
                d[nextNode] = weight + dist;
                pq.push({d[nextNode], nextNode});
            }
        }
        return d;
    }
```

BellmanFord.h

Description: Calculates shortest paths from s in a graph that might have negative edge weights. Unreachable nodes get $\text{dist} = \text{inf}$; nodes reachable through negative-weight cycles get $\text{dist} = -\text{inf}$. Assumes $V^2 \max |w_i| < \sim 2^{83}$.
Time: $\mathcal{O}(VE)$

830a8f, 23 lines

```
const ll inf = LLONG_MAX;
struct Ed { int a, b, w, s() { return a < b ? a : -a; }};
struct Node { ll dist = inf; int prev = -1; };

void bellmanFord(vector<Node>& nodes, vector<Ed>& eds, int s) {
    nodes[s].dist = 0;
    sort(all(eds), [](Ed a, Ed b) { return a.s() < b.s(); });

    int lim = sz(nodes) / 2 + 2; // /3+100 with shuffled vertices
    rep(i,0,lim) for (Ed ed : eds) {
        Node cur = nodes[ed.a], &dest = nodes[ed.b];
        if (abs(cur.dist) == inf) continue;
        ll d = cur.dist + ed.w;
        if (d < dest.dist) {
            dest.prev = ed.a;
```

```
        dest.dist = (i < lim-1 ? d : -inf);
    }
}
rep(i,0,lim) for (Ed e : eds) {
    if (nodes[e.a].dist == -inf)
        nodes[e.b].dist = -inf;
}
}
```

FloydWarshall.h

Description: Calculates all-pairs shortest path in a directed graph that might have negative edge weights. Input is an distance matrix m , where $m[i][j] = \text{inf}$ if i and j are not adjacent. As output, $m[i][j]$ is set to the shortest distance between i and j , inf if no path, or $-\text{inf}$ if the path goes through a negative-weight cycle.
Time: $\mathcal{O}(N^3)$

531245, 12 lines

```
const ll inf = 1LL << 62;
void floydWarshall(vector<vector<ll>>& m) {
    int n = sz(m);
    rep(i,0,n) m[i][i] = min(m[i][i], 0LL);
    rep(k,0,n) rep(i,0,n) rep(j,0,n)
        if (m[i][k] != inf && m[k][j] != inf) {
            auto newDist = max(m[i][k] + m[k][j], -inf);
            m[i][j] = min(m[i][j], newDist);
        }
    rep(k,0,n) if (m[k][k] < 0) rep(i,0,n) rep(j,0,n)
        if (m[i][k] != inf && m[k][j] != inf) m[i][j] = -inf;
}
```

TopoSort.h

Description: Topological sorting. Given is an oriented graph. Output is an ordering of vertices, such that there are edges only from left to right. If there are cycles, the returned list will have size smaller than n – nodes reachable from cycles will not be returned.
Time: $\mathcal{O}(|V| + |E|)$

d678d8, 8 lines

```
vi topoSort(const vector<vi>& gr) {
    vi indeg(sz(gr)), q;
    for (auto& li : gr) for (int x : li) indeg[x]++;
    rep(i,0,sz(gr)) if (indeg[i] == 0) q.push_back(i);
    rep(j,0,sz(q)) for (int x : gr[q[j]])
        if (--indeg[x] == 0) q.push_back(x);
    return q;
}
```

6.2 Network flow

PushRelabel.h

Description: Push-relabel using the highest label selection rule and the gap heuristic. Quite fast in practice. To obtain the actual flow, look at positive values only.

Time: $\mathcal{O}\left(V^2\sqrt{E}\right)$

0ae1d4, 48 lines

```
struct PushRelabel {
    struct Edge {
        int dest, back;
        ll f, c;
    };
    vector<vector<Edge>> g;
    vector<ll> ec;
    vector<Edge*> cur;
    vector<vi> hs; vi H;
    PushRelabel(int n) : g(n), ec(n), cur(n), hs(2*n), H(n) {}

    void addEdge(int s, int t, ll cap, ll rcap=0) {
        if (s == t) return;
        g[s].push_back({t, sz(g[t]), 0, cap});
        g[t].push_back({s, sz(g[s])-1, 0, rcap});
```

```
    }

    void addFlow(Edge& e, ll f) {
        Edge &back = g[e.dest][e.back];
        if (!ec[e.dest] && f) hs[H[e.dest]].push_back(e.dest);
        e.f += f; e.c -= f; ec[e.dest] += f;
        back.f -= f; back.c += f; ec[back.dest] -= f;
    }
    ll calc(int s, int t) {
        int v = sz(g); H[s] = v; ec[t] = 1;
        vi co(2*v); co[0] = v-1;
        rep(i,0,v) cur[i] = g[i].data();
        for (Edge& e : g[s]) addFlow(e, e.c);

        for (int hi = 0;;) {
            while (hs[hi].empty()) if (!hi--) return -ec[s];
            int u = hs[hi].back(); hs[hi].pop_back();
            while (ec[u] > 0) // discharge u
                if (cur[u] == g[u].data() + sz(g[u])) {
                    H[u] = 1e9;
                    for (Edge& e : g[u]) if (e.c && H[u] > H[e.dest]+1)
                        H[u] = H[e.dest]+1, cur[u] = &e;
                    if (++co[H[u]],!--co[hi] && hi < v)
                        rep(i,0,v) if (hi < H[i] && H[i] < v)
                            --co[H[i]], H[i] = v + 1;
                    hi = H[u];
                } else if (cur[u]->c && H[u] == H[cur[u]->dest]+1)
                    addFlow(*cur[u], min(ec[u], cur[u]->c));
                else ++cur[u];
            }
        }
        bool leftOfMinCut(int a) { return H[a] >= sz(g); }
    };
};
```

MinCostMaxFlow.h

Description: Min-cost max-flow. If costs can be negative, call `setpi` before `maxflow`, but note that negative cost cycles are not supported. To obtain the actual flow, look at positive values only.

Time: $\mathcal{O}(FE \log(V))$ where F is max flow. $\mathcal{O}(VE)$ for `setpi`.

58385b, 79 lines

```
#include <bits/extc++.h>

const ll INF = numeric_limits<ll>::max() / 4;

struct MCMF {
    struct edge {
        int from, to, rev;
        ll cap, cost, flow;
    };
    int N;
    vector<vector<edge>> ed;
    vi seen;
    vector<ll> dist, pi;
    vector<edge*> par;

    MCMF(int N) : N(N), ed(N), seen(N), dist(N), pi(N), par(N) {}

    void addEdge(int from, int to, ll cap, ll cost) {
        if (from == to) return;
        ed[from].push_back(edge{ from,to,sz(ed[to]),cap,cost,0 });
        ed[to].push_back(edge{ to,from,sz(ed[from])-1,0,-cost,0 });
    }

    void path(int s) {
        fill(all(seen), 0);
        fill(all(dist), INF);
        dist[s] = 0; ll di;

        __gnu_pbds::priority_queue<pair<ll, int>> q;
```

```
vector<decltype(q)::point_iterator> its(N);
q.push({ 0, s });

while (!q.empty()) {
    s = q.top().second; q.pop();
    seen[s] = 1; di = dist[s] + pi[s];
    for (edge& e : ed[s]) if (!seen[e.to]) {
        ll val = di - pi[e.to] + e.cost;
        if (e.cap - e.flow > 0 && val < dist[e.to]) {
            dist[e.to] = val;
            par[e.to] = &e;
            if (its[e.to] == q.end())
                its[e.to] = q.push({ -dist[e.to], e.to });
            else
                q.modify(its[e.to], { -dist[e.to], e.to });
        }
    }
}
rep(i,0,N) pi[i] = min(pi[i] + dist[i], INF);
}

pair<ll, ll> maxflow(int s, int t) {
    ll totflow = 0, totcost = 0;
    while (path(s), seen[t]) {
        ll fl = INF;
        for (edge* x = par[t]; x; x = par[x->from])
            fl = min(fl, x->cap - x->flow);

        totflow += fl;
        for (edge* x = par[t]; x; x = par[x->from]) {
            x->flow += fl;
            ed[x->to][x->rev].flow -= fl;
        }
    }
    rep(i,0,N) for(edge& e : ed[i]) totcost += e.cost * e.flow;
    return {totflow, totcost/2};
}

// If some costs can be negative, call this before maxflow:
void setpi(int s) { // (otherwise, leave this out)
    fill(all(pi), INF); pi[s] = 0;
    int it = N, ch = 1; ll v;
    while (ch-- && it--)
        rep(i,0,N) if (pi[i] != INF)
            for (edge& e : ed[i]) if (e.cap)
                if ((v = pi[i] + e.cost) < pi[e.to])
                    pi[e.to] = v, ch = 1;
    assert(it >= 0); // negative cost cycle
}
};
```

EdmondsKarp.h

Description: Flow algorithm with guaranteed complexity $O(VE^2)$. To get edge flow values, compare capacities before and after, and take the positive values only.

<pre>template<class T> T edmondsKarp(vector<unordered_map<int, T>&& graph, int source, int sink) { assert(source != sink); T flow = 0; vi par(sz(graph)), q = par; for (;;) { fill(all(par), -1); par[source] = 0; int ptr = 1; q[0] = source; rep(i,0,ptr) {</pre>	482fe0, 36 lines
--	------------------

```
int x = q[i];
for (auto e : graph[x]) {
    if (par[e.first] == -1 && e.second > 0) {
        par[e.first] = x;
        q[ptr++] = e.first;
        if (e.first == sink) goto out;
    }
}
}
return flow;
out:
T inc = numeric_limits<T>::max();
for (int y = sink; y != source; y = par[y])
    inc = min(inc, graph[par[y]][y]);

flow += inc;
for (int y = sink; y != source; y = par[y]) {
    int p = par[y];
    if ((graph[p][y] -= inc) <= 0) graph[p].erase(y);
    graph[y][p] += inc;
}
}
}
```

MinCut.h

Description: After running max-flow, the left side of a min-cut from s to t is given by all vertices reachable from s , only traversing edges with positive residual capacity.

GlobalMinCut.h

Description: Find a global minimum cut in an undirected graph, as represented by an adjacency matrix.
Time: $\mathcal{O}(V^3)$

<pre>pair<int, vi> globalMinCut(vector<vi> mat) { pair<int, vi> best = {INT_MAX, {}}; int n = sz(mat); vector<vi> co(n); rep(i,0,n) co[i] = {i}; rep(ph,1,n) { vi w = mat[0]; size_t s = 0, t = 0; rep(it,0,n-ph) { // $O(V^2) \rightarrow O(E \log V)$ with prio. queue w[t] = INT_MIN; s = t, t = max_element(all(w)) - w.begin(); rep(i,0,n) w[i] += mat[t][i]; } best = min(best, {w[t] - mat[t][t], co[t]}); co[s].insert(co[s].end(), all(co[t])); rep(i,0,n) mat[s][i] += mat[t][i]; rep(i,0,n) mat[i][s] = mat[s][i]; mat[0][t] = INT_MIN; } return best; }</pre>	8b0e19, 21 lines
--	------------------

GomoryHu.h

Description: Given a list of edges representing an undirected flow graph, returns edges of the Gomory-Hu tree. The max flow between any pair of vertices is given by minimum edge weight along the Gomory-Hu tree path.
Time: $\mathcal{O}(V)$ Flow Computations

<pre>"PushRelabel.h" typedef array<ll, 3> Edge; vector<Edge> gomoryHu(int N, vector<Edge> ed) { vector<Edge> tree; vi par(N); rep(i,1,N) {</pre>	0418b3, 13 lines
---	------------------

```
PushRelabel D(N); // Dinic also works
for (Edge t : ed) D.addEdge(t[0], t[1], t[2], t[2]);
tree.push_back({i, par[i], D.calc(i, par[i])});
rep(j,i+1,N)
    if (par[j] == par[i] && D.leftOfMinCut(j)) par[j] = i;
}
return tree;
}
```

6.3 Matching

hopcroftKarp.h

Description: Fast bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and $btoa$ should be a vector full of -1's of the same size as the right partition. Returns the size of the matching. $btoa[i]$ will be the match for vertex i on the right side, or -1 if it's not matched.

Usage: vi btoa(m, -1); hopcroftKarp(g, btoa);
Time: $\mathcal{O}(\sqrt{VE})$

<pre>bool dfs(int a, int L, vector<vi>& g, vi& btoa, vi& A, vi& B) { if (A[a] != L) return 0; A[a] = -1; for (int b : g[a]) if (B[b] == L + 1) { B[b] = 0; if (btoa[b] == -1 dfs(btoa[b], L + 1, g, btoa, A, B)) return btoa[b] = a, 1; } return 0; }</pre>	f612e4, 42 lines
--	------------------

```
int hopcroftKarp(vector<vi>& g, vi& btoa) {
    int res = 0;
    vi A(g.size()), B(btoa.size()), cur, next;
    for (;;) {
        fill(all(A), 0);
        fill(all(B), 0);
        cur.clear();
        for (int a : btoa) if(a != -1) A[a] = -1;
        rep(a,0,sz(g)) if(A[a] == 0) cur.push_back(a);
        for (int lay = 1;; lay++) {
            bool islast = 0;
            next.clear();
            for (int a : cur) for (int b : g[a]) {
                if (btoa[b] == -1) {
                    B[b] = lay;
                    islast = 1;
                }
                else if (btoa[b] != a && !B[b]) {
                    B[b] = lay;
                    next.push_back(btoa[b]);
                }
            }
            if (islast) break;
            if (next.empty()) return res;
            for (int a : next) A[a] = lay;
            cur.swap(next);
        }
        rep(a,0,sz(g))
            res += dfs(a, 0, g, btoa, A, B);
    }
}
```

DFSMatching.h

Description: Simple bipartite matching algorithm. Graph g should be a list of neighbors of the left partition, and $btoa$ should be a vector full of -1's of the same size as the right partition. Returns the size of the matching. $btoa[i]$ will be the match for vertex i on the right side, or -1 if it's not matched.

Usage: vi btoa(m, -1); dfsMatching(g, btoa);
Time: $\mathcal{O}(VE)$

```
bool find(int j, vector<vi>& g, vi& btoa, vi& vis) {
    if (btoa[j] == -1) return 1;
    vis[j] = 1; int di = btoa[j];
    for (int e : g[di])
        if (!vis[e] && find(e, g, btoa, vis)) {
            btoa[e] = di;
            return 1;
        }
    return 0;
}
int dfsMatching(vector<vi>& g, vi& btoa) {
    vi vis;
    rep(i,0,sz(g)) {
        vis.assign(sz(btoa), 0);
        for (int j : g[i])
            if (find(j, g, btoa, vis)) {
                btoa[j] = i;
                break;
            }
    }
    return sz(btoa) - (int)count(all(btoa), -1);
}
```

MinimumVertexCover.h

Description: Finds a minimum vertex cover in a bipartite graph. The size is the same as the size of a maximum matching, and the complement is a maximum independent set.

"DFSMatching.h"	da4196, 20 lines
-----------------	------------------

```
vi cover(vector<vi>& g, int n, int m) {
    vi match(m, -1);
    int res = dfsMatching(g, match);
    vector<bool> lfound(n, true), seen(m);
    for (int it : match) if (it != -1) lfound[it] = false;
    vi q, cover;
    rep(i,0,n) if (lfound[i]) q.push_back(i);
    while (!q.empty()) {
        int i = q.back(); q.pop_back();
        lfound[i] = 1;
        for (int e : g[i]) if (!seen[e] && match[e] != -1) {
            seen[e] = true;
            q.push_back(match[e]);
        }
    }
    rep(i,0,n) if (!lfound[i]) cover.push_back(i);
    rep(i,0,m) if (seen[i]) cover.push_back(n+i);
    assert(sz(cover) == res);
    return cover;
}
```

WeightedMatching.h

Description: Given a weighted bipartite graph, matches every node on the left with a node on the right such that no nodes are in two matchings and the sum of the edge weights is minimal. Takes cost[N][M], where cost[i][j] = cost for L[i] to be matched with R[j] and returns (min cost, match), where L[i] is matched with R[match[i]]. Negate costs for max cost. Requires $N \leq M$.

"Time: $\mathcal{O}(N^2M)$ "	1e0fe9, 31 lines
------------------------------	------------------

```
pair<int, vi> hungarian(const vector<vi> &a) {
    if (a.empty()) return {0, {}};
    int n = sz(a) + 1, m = sz(a[0]) + 1;
    vi u(n), v(m), p(m), ans(n - 1);
    rep(i,1,n) {
        p[0] = i;
        int j0 = 0; // add "dummy" worker 0
        vi dist(m, INT_MAX), pre(m, -1);
        vector<bool> done(m + 1);
        do { // dijkstra
            done[j0] = true;
```

```
int i0 = p[j0], j1, delta = INT_MAX;
rep(j,1,m) if (!done[j]) {
    auto cur = a[i0 - 1][j - 1] - u[i0] - v[j];
    if (cur < dist[j]) dist[j] = cur, pre[j] = j0;
    if (dist[j] < delta) delta = dist[j], j1 = j;
}
rep(j,0,m) {
    if (done[j]) u[p[j]] += delta, v[j] -= delta;
    else dist[j] -= delta;
}
j0 = j1;
} while (p[j0]);
while (j0) { // update alternating path
    int j1 = pre[j0];
    p[j0] = p[j1], j0 = j1;
}
}
rep(j,1,m) if (p[j]) ans[p[j] - 1] = j - 1;
return {-v[0], ans}; // min cost
}
```

GeneralMatching.h

Description: Matching for general graphs. Fails with probability N/mod . **Time:** $\mathcal{O}(N^3)$

"../numerical/MatrixInverse-mod.h"	cb1912, 40 lines
------------------------------------	------------------

```
vector<pii> generalMatching(int N, vector<pii>& ed) {
    vector<vector<ll>> mat(N, vector<ll>(N)), A;
    for (pii pa : ed) {
        int a = pa.first, b = pa.second, r = rand() % mod;
        mat[a][b] = r, mat[b][a] = (mod - r) % mod;
    }

    int r = matInv(A = mat), M = 2*N - r, fi, fj;
    assert(r % 2 == 0);

    if (M != N) do {
        mat.resize(M, vector<ll>(M));
        rep(i,0,N) {
            mat[i].resize(M);
            rep(j,N,M) {
                int r = rand() % mod;
                mat[i][j] = r, mat[j][i] = (mod - r) % mod;
            }
        }
    } while (matInv(A = mat) != M);

    vi has(M, 1); vector<pii> ret;
    rep(it,0,M/2) {
        rep(i,0,M) if (has[i])
            rep(j,i+1,M) if (A[i][j] && mat[i][j]) {
                fi = i; fj = j; goto done;
            } assert(0); done:
        if (fj < N) ret.emplace_back(fi, fj);
        has[fi] = has[fj] = 0;
        rep(sw,0,2) {
            ll a = modpow(A[fi][fj], mod-2);
            rep(i,0,M) if (has[i] && A[i][fj]) {
                ll b = A[i][fj] * a % mod;
                rep(j,0,M) A[i][j] = (A[i][j] - A[fi][j] * b) % mod;
            }
            swap(fi,fj);
        }
    }
    return ret;
}
```

6.4 DFS algorithms

SCC.h

Description: ... **Time:** ...

"f1b591, 23 lines"

```
vvll gr(N), revgr(N);
vector<bool> vis(N);
stack<ll> st;
void dfs1(ll node) {
    vis[node] = 1;
    for (ll ch : gr[node]) { if (!vis[ch]) dfs1(ch); }
    st.push(node);
}
void dfs2(ll node, vll &comp) {
    vis[node] = 1; comp.push_back(node);
    for (ll ch : revgr[node]) { if (!vis[ch]) dfs2(ch, comp); }
}
vvll kosaraju() { // returns comps in topological order
    vvll comps;
    L(i, 1, n) if (!vis[i]) dfs1(i);
    L(i, 1, n) vis[i] = 0;
    while (st.size()) {
        ll now = st.top(); st.pop();
        if (!vis[now]) {
            vll comp; dfs2(now, comp); comps.push_back(comp);
        }
    }
    return comps;
}
```

BiconnectedComponents.h

Description: Finds all biconnected components in an undirected graph, and runs a callback for the edges in each. In a biconnected component there are at least two internally disjoint paths between any two nodes (a cycle exists through them). Note that a node can be in several components. An edge which is not in a component is a bridge, i.e., not part of any cycle.

Usage: int eid = 0; ed.resize(N);
for each edge (a,b) {
ed[a].emplace_back(b, eid);
ed[b].emplace_back(a, eid++);
}
bicomps([&](const vi& edgelist) {...});
Time: $\mathcal{O}(E + V)$

"c6b7c7, 32 lines"

```
vi num, st;
vector<vector<pii>> ed;
int Time;
template<class F>
int dfs(int at, int par, F& f) {
    int me = num[at] = ++Time, top = me;
    for (auto [y, e] : ed[at]) if (e != par) {
        if (num[y]) {
            top = min(top, num[y]);
            if (num[y] < me)
                st.push_back(e);
        } else {
            int si = sz(st);
            int up = dfs(y, e, f);
            top = min(top, up);
            if (up == me) {
                st.push_back(e);
                f(vi(st.begin() + si, st.end()));
                st.resize(si);
            }
            else if (up < me) st.push_back(e);
            else { /* e is a bridge */ }
        }
    }
    return top;
}
```

```

template<class F>
void bicomps(F f) {
    num.assign(sz(ed), 0);
    rep(i,0,sz(ed)) if (!num[i]) dfs(i, -1, f);
}

```

2sat.h

Description: Calculates a valid assignment to boolean variables a, b, c,... to a 2-SAT problem, so that an expression of the type $(a||b)\&\&(!a||c)\&\&(d||b)\&\&...$ becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions (~x).
Usage: TwoSat ts(number of boolean variables);
ts.either(0, ~3); // Var 0 is true or var 3 is false
ts.setValue(2); // Var 2 is true
ts.atMostOne({0,~1,2}); // <= 1 of vars 0, ~1 and 2 are true
ts.solve(); // Returns true iff it is solvable
ts.values[0..N-1] holds the assigned values to the vars
Time: $\mathcal{O}(N + E)$, where N is the number of boolean variables, and E is the number of clauses.

484741, 56 lines

```

// variables are zero indexed
struct twosat{
    int n;
    vector<vector<int>> g,gt;
    vector<bool> vis,res;
    vector<int> comp;
    stack<int> ts;
    twosat(int vars=0){
        n=vars<<1; g.resize(n); gt.resize(n); }

    //if you want to force variable a to be true : addOR (a,1,a,1)
    ;
    //if you want to force variable a to be false : addOR (a,0,a,0);

```

```

// adds an implication
void _add(int a,bool sign_a,int b,bool sign_b){
    a+=a+(sign_a^1);
    b+=b+(sign_b^1);
    g[a].push_back(b); gt[b].push_back(a); }

//(x_a or (not x_b)-> sign_a=1,sign_b=0
void addOR(int a,bool sign_a,int b,bool sign_b){
    a+=a+(sign_a^1);
    b+=b+(sign_b^1);
    g[a^1].push_back(b); // !a => b
    g[b^1].push_back(a); // !b => a
    gt[b].push_back(a^1);
    gt[a].push_back(b^1); }

```

```

//(!x_a xor !x_b)-> sign_a=0, sign_b=0
void addXOR(int a,bool sign_a,int b,bool sign_b){
    addOR(a,sign_a,b,sign_b);
    addOR(a,!sign_a,b,!sign_b); }

void dfs1(int u){
    vis[u]=true;
    for(int v:g[u]) if(!vis[v]) dfs1(v);
    ts.push(u); }

void dfs2(int u,int c){
    comp[u]=c;
    for(int v:gt[u]) if(comp[v]==-1) dfs2(v,c); }

```

```

bool run(){ // returns true if possible, else false
    vis.resize(n,false);
    for(int i=0;i<n;++i) if(!vis[i]) dfs1(i);
    int scc=0;
    comp.resize(n,-1);

```

2sat EulerWalk Bridges EulerCycle EdgeColoring

```

while(!ts.empty()){
    int u=ts.top(); ts.pop();
    if(comp[u]==-1) dfs2(u,scc++); }
res.resize(n/2);
for(int i=0;i<n;i+=2){
    if(comp[i]==comp[i+1]) return false;
    res[i/2]=(comp[i]>comp[i+1]); }
return true; }
};

```

EulerWalk.h

Description: Eulerian undirected/directed path/cycle algorithm. Input should be a vector of (dest, global edge index), where for undirected graphs, forward/backward edges have the same index. Returns a list of nodes in the Eulerian path/cycle with src at both start and end, or empty list if no cycle/path exists. To get edge indices back, add .second to s and ret.
Time: $\mathcal{O}(V + E)$

780b64, 15 lines

```

vi eulerWalk(vector<vector<pii>>& gr, int nedges, int src=0) {
    int n = sz(gr);
    vi D(n), its(n), eu(nedges), ret, s = {src};
    D[src]++; // to allow Euler paths, not just cycles
    while (!s.empty()) {
        int x = s.back(), y, e, &it = its[x], end = sz(gr[x]);
        if (it == end){ ret.push_back(x); s.pop_back(); continue; }
        tie(y, e) = gr[x][it++];
        if (!eu[e]) {
            D[x]--, D[y]++;
            eu[e] = 1; s.push_back(y);
        }
    }
    for (int x : D) if (x < 0 || sz(ret) != nedges+1) return {};
    return {ret.rbegin(), ret.rend()};
}

```

Bridges.h

Description: ...
Time: ...

98ffe3, 36 lines

```

struct BridgesAndCuts{
    int n,timer=0;
    vector<vector<int>> g;
    vector<int> tin,low;
    vector<char> vis,is_cut;
    vector<pair<int,int>> bridges;

```

```

    BridgesAndCuts(int n=0):n(n),g(n),tin(n,-1),
        low(n,-1),vis(n,0),is_cut(n,0){}

    void addEdge(int u,int v){ g[u].push_back(v); g[v].push_back(u); }

    void run(){
        timer=0;
        fill(tin.begin(),tin.end(),-1);
        fill(low.begin(),low.end(),-1);
        fill(vis.begin(),vis.end(),0);
        fill(is_cut.begin(),is_cut.end(),0);
        bridges.clear();
        for(int v=0;v<n;+v) if(!vis[v]) dfs(v,-1); }

```

```

private:
void dfs(int v,int p){
    vis[v]=1;
    tin[v]=low[v]=timer++;
    int children=0;
    for(int to:g[v]){
        if(to==p) continue;
        if(vis[to]){ low[v]=min(low[v],tin[to]); }
        else{

```

```

        ++children; dfs(to,v);
        low[v]=min(low[v],low[to]);
        if(low[to]>tin[v]) bridges.emplace_back(min(v,to),max(v,to));
        };
        if(p!=-1&&low[to]>=tin[v]) is_cut[v]=1; } }
    if(p==-1&&children>1) is_cut[v]=1; }
};

```

EulerCycle.h

Description: Order statistics
Usage: ...
Time: ...

92d8fa, 22 lines

```

vector<ar<ll, 2>> gr[N];
vector<bool> vis_edge(N);
vll ans;
void dfs(ll node) {
    while(!gr[node].empty()) {
        auto [ch, ed] = gr[node].back(); gr[node].pop_back();
        if(vis_edge[ed]) continue;
        vis_edge[ed] = 1;
        dfs(ch);
    }
    ans.push_back(node);
}
main:
L(i, 1, m) {
    cin >> u >> v;
    gr[u].push_back({v, i}); gr[v].push_back({u, i});
    deg[u]++, deg[v]++;
}
L(i, 1, n) if(deg[i]&1) {cout << "IMPOSSIBLE" << nl; return;}
dfs(1);
if(sz(ans)!=m+1) {cout << "IMPOSSIBLE" << nl; return;}
for(ll x : ans) cout << x << gp;

```

6.5 Coloring

EdgeColoring.h

Description: Given a simple, undirected graph with max degree D , computes a $(D + 1)$ -coloring of the edges such that no neighboring edges share a color. (D -coloring is NP-hard, but can be done for bipartite graphs by repeated matchings of max-degree nodes.)
Time: $\mathcal{O}(NM)$

e210c2, 31 lines

```

vi edgeColoring(int N, vector<pii> eds) {
    vi cc(N + 1), ret(sz(eds)), fan(N), free(N), loc;
    for (pii e : eds) ++cc[e.first], ++cc[e.second];
    int u, v, ncols = *max_element(all(cc)) + 1;
    vector<vi> adj(N, vi(ncols, -1));
    for (pii e : eds) {
        tie(u, v) = e;
        fan[0] = v;
        loc.assign(ncols, 0);
        int at = u, end = u, d, c = free[u], ind = 0, i = 0;
        while (d = free[v], !loc[d] && (v = adj[u][d]) != -1)
            loc[d] = ++ind, cc[ind] = d, fan[ind] = v;
        cc[loc[d]] = c;
        for (int cd = d; at != -1; cd ^= c ^ d, at = adj[at][cd])
            swap(adj[at][cd], adj[end = at][cd ^ c ^ d]);
        while (adj[fan[i]][d] != -1) {
            int left = fan[i], right = fan[++i], e = cc[i];
            adj[u][e] = left;
            adj[left][e] = u;
            adj[right][e] = -1;
            free[right] = e;
        }
        adj[u][d] = fan[i];
        adj[fan[i]][d] = u;
        for (int y : {fan[0], u, end})

```

```
    for (int& z = free[y] = 0; adj[y][z] != -1; z++);
}
rep(i,0,sz(eds))
    for (tie(u, v) = eds[i]; adj[u][ret[i]] != v;) ++ret[i];
return ret;
}
```

6.6 Heuristics

MaximalCliques.h

Description: Runs a callback for all maximal cliques in a graph (given as a symmetric bitset matrix; self-edges not allowed). Callback is given a bitset representing the maximal clique.

Time: $\mathcal{O}\left(3^{n/3}\right)$, much faster for sparse graphs

```
typedef bitset<128> B;
template<class F>
void cliques(vector<B>& eds, F f, B P = ~B(), B X={}, B R={}) {
    if (!P.any()) { if (!X.any()) f(R); return; }
    auto q = (P | X)._Find_first();
    auto cands = P & ~eds[q];
    rep(i,0,sz(eds)) if (cands[i]) {
        R[i] = 1;
        cliques(eds, f, P & eds[i], X & eds[i], R);
        R[i] = P[i] = 0; X[i] = 1;
    }
}
```

MaximumClique.h

Description: Quickly finds a maximum clique of a graph (given as symmetric bitset matrix; self-edges not allowed). Can be used to find a maximum independent set by finding a clique of the complement graph.

Time: Runs in about 1s for n=155 and worst case random graphs (p=.90). Runs faster for sparse graphs.

```
typedef vector<bitset<200>> vb;
struct MaxClique {
    double limit=0.025, pk=0;
    struct Vertex { int i, d=0; };
    typedef vector<Vertex> vv;
    vb e;
    vv V;
    vector<vi> C;
    vi qmax, q, S, old;
    void init(vv& r) {
        for (auto& v : r) v.d = 0;
        for (auto& v : r) for (auto j : r) v.d += e[v.i][j.i];
        sort(all(r), [](auto a, auto b) { return a.d > b.d; });
        int mxD = r[0].d;
        rep(i,0,sz(r)) r[i].d = min(i, mxD) + 1;
    }
    void expand(vv& R, int lev = 1) {
        S[lev] += S[lev - 1] - old[lev];
        old[lev] = S[lev - 1];
        while (sz(R)) {
            if (sz(q) + R.back().d <= sz(qmax)) return;
            q.push_back(R.back().i);
            vv T;
            for(auto v:R) if (e[R.back().i][v.i]) T.push_back({v.i});
            if (sz(T)) {
                if (S[lev]++ / ++pk < limit) init(T);
                int j = 0, mxk = 1, mnk = max(sz(qmax) - sz(q) + 1, 1);
                C[1].clear(), C[2].clear();
                for (auto v : T) {
                    int k = 1;
                    auto f = [&](int i) { return e[v.i][i]; };
                    while (any_of(all(C[k]), f)) k++;
                    if (k > mxk) mxk = k, C[mxk + 1].clear();
                    if (k < mnk) T[j++].i = v.i;
                }
            }
        }
    }
};
```

```
    C[k].push_back(v.i);
}
if (j > 0) T[j - 1].d = 0;
rep(k,mnk,mxk + 1) for (int i : C[k])
    T[j].i = i, T[j++].d = k;
expand(T, lev + 1);
} else if (sz(q) > sz(qmax)) qmax = q;
q.pop_back(), R.pop_back();
}
}
vi maxClique() { init(V), expand(V); return qmax; }
MaxClique(vb conn) : e(conn), C(sz(e)+1), S(sz(C)), old(S) {
    rep(i,0,sz(e)) V.push_back({i});
}
};
```

MaximumIndependentSet.h

Description: To obtain a maximum independent set of a graph, find a max clique of the complement. If the graph is bipartite, see MinimumVertex-Cover.

6.7 Trees

BinaryLifting.h

Description: Calculate power of two jumps in a tree, to support fast upward jumps and LCAs. Assumes the root node points to itself.

Time: construction $\mathcal{O}(N \log N)$, queries $\mathcal{O}(\log N)$

```
vector<vi> treeJump(vi& P){
    int on = 1, d = 1;
    while(on < sz(P)) on *= 2, d++;
    vector<vi> jmp(d, P);
    rep(i,1,d) rep(j,0,sz(P))
        jmp[i][j] = jmp[i-1][jmp[i-1][j]];
    return jmp;
}
```

```
int jmp(vector<vi>& tbl, int nod, int steps){
    rep(i,0,sz(tbl))
        if(steps&(1<<i)) nod = tbl[i][nod];
    return nod;
}
```

```
int lca(vector<vi>& tbl, vi& depth, int a, int b) {
    if (depth[a] < depth[b]) swap(a, b);
    a = jmp(tbl, a, depth[a] - depth[b]);
    if (a == b) return a;
    for (int i = sz(tbl); i--;) {
        int c = tbl[i][a], d = tbl[i][b];
        if (c != d) a = c, b = d;
    }
    return tbl[0][a];
}
```

LCA.h

Description: Data structure for computing lowest common ancestors in a tree (with 0 as root). C should be an adjacency list of the tree, either directed or undirected.

Time: $\mathcal{O}(N \log N + Q)$

```
"/>
```

```

    }
}
void dfsHld(int v) {
    pos[v] = tim++;
    for (int u : adj[v]) {
        rt[u] = (u == adj[v][0] ? rt[v] : u);
        dfsHld(u);
    }
}
template <class B> void process(int u, int v, B op) {
    for (; v = par[rt[v]]) {
        if (pos[u] > pos[v]) swap(u, v);
        if (rt[u] == rt[v]) break;
        op(pos[rt[v]], pos[v] + 1);
    }
    op(pos[u] + VALS_EDGES, pos[v] + 1);
}
void modifyPath(int u, int v, int val) {
    process(u, v, [&](int l, int r) { tree->add(l, r, val); });
}
void modifySubtree(int v, ll val) {
    tree->add(pos[v] + VALS_EDGES, pos[v] + siz[v], val);
}
int queryPath(int u, int v) { // Modify depending on problem
    int res = -1e9;
    process(u, v, [&](int l, int r) {
        res = max(res, (tree->query(l, r)).mx );
    });
    return res;
}
int querySubtree(int v) { // modifySubtree is similar
    return (tree->query(pos[v] + VALS_EDGES, pos[v] + siz[v])).mx;
}
};

```

LinkCutTree.h

Description: Represents a forest of unrooted trees. You can add and remove edges (as long as the result is still a forest), and check whether two nodes are in the same tree.

Time: All operations take amortized $\mathcal{O}(\log N)$.

0fb462, 90 lines

```

struct Node { // Splay tree. Root's pp contains tree's parent.
    Node *p = 0, *pp = 0, *c[2];
    bool flip = 0;
    Node() { c[0] = c[1] = 0; fix(); }
    void fix() {
        if (c[0]) c[0]->p = this;
        if (c[1]) c[1]->p = this;
        // (+ update sum of subtree elements etc. if wanted)
    }
    void pushFlip() {
        if (!flip) return;
        flip = 0; swap(c[0], c[1]);
        if (c[0]) c[0]->flip ^= 1;
        if (c[1]) c[1]->flip ^= 1;
    }
    int up() { return p ? p->c[1] == this : -1; }
    void rot(int i, int b) {
        int h = i ^ b;
        Node *x = c[i], *y = b == 2 ? x : x->c[h], *z = b ? y : x;
        if ((y->p = p)) p->c[up()] = y;
        c[i] = z->c[i ^ 1];
        if (b < 2) {
            x->c[h] = y->c[h ^ 1];
            y->c[h ^ 1] = x;
        }
        z->c[i ^ 1] = this;
        fix(); x->fix(); y->fix();
    }
};

```

```

    if (p) p->fix();
    swap(pp, y->pp);
}
void splay() {
    for (pushFlip(); p; ) {
        if (p->p) p->p->pushFlip();
        p->pushFlip(); pushFlip();
        int c1 = up(), c2 = p->up();
        if (c2 == -1) p->rot(c1, 2);
        else p->p->rot(c2, c1 != c2);
    }
}
Node* first() {
    pushFlip();
    return c[0] ? c[0]->first() : (splay(), this);
}
};

```

```

struct LinkCut {
    vector<Node> node;
    LinkCut(int N) : node(N) {}

    void link(int u, int v) { // add an edge (u, v)
        assert(!connected(u, v));
        makeRoot(&node[u]);
        node[u].pp = &node[v];
    }
    void cut(int u, int v) { // remove an edge (u, v)
        Node *x = &node[u], *top = &node[v];
        makeRoot(top); x->splay();
        assert(top == (x->pp ? x->c[0]));
        if (x->pp) x->pp = 0;
        else {
            x->c[0] = top->p = 0;
            x->fix();
        }
    }
    bool connected(int u, int v) { // are u, v in the same tree?
        Node* nu = access(&node[u])->first();
        return nu == access(&node[v])->first();
    }
    void makeRoot(Node* u) {
        access(u);
        u->splay();
        if (u->c[0]) {
            u->c[0]->p = 0;
            u->c[0]->flip ^= 1;
            u->c[0]->pp = u;
            u->c[0] = 0;
            u->fix();
        }
    }
    Node* access(Node* u) {
        u->splay();
        while (Node* pp = u->pp) {
            pp->splay(); u->pp = 0;
            if (pp->c[1]) {
                pp->c[1]->p = 0; pp->c[1]->pp = pp; }
            pp->c[1] = u; pp->fix(); u = pp;
        }
        return u;
    }
};

```

DirectedMST.h

Description: Finds a minimum spanning tree/arborescence of a directed graph, given a root node. If no MST exists, returns -1.

Time: $\mathcal{O}(E \log V)$

"../data-structures/UnionFindRollback.h"

```

struct Edge { int a, b; ll w; };
struct Node {
    Edge key;
    Node *l, *r;
    ll delta;
    void prop() {
        key.w += delta;
        if (l) l->delta += delta;
        if (r) r->delta += delta;
        delta = 0;
    }
    Edge top() { prop(); return key; }
};
Node *merge(Node *a, Node *b) {
    if (!a || !b) return a ? b;
    a->prop(), b->prop();
    if (a->key.w > b->key.w) swap(a, b);
    swap(a->l, (a->r = merge(b, a->r)));
    return a;
}
void pop(Node& a) { a->prop(); a = merge(a->l, a->r); }

```

```

pair<ll, vi> dmst(int n, int r, vector<Edge>& g) {
    RollbackUF uf(n);
    vector<Node*> heap(n);
    for (Edge e : g) heap[e.b] = merge(heap[e.b], new Node(e));
    ll res = 0;
    vi seen(n, -1), path(n), par(n);
    seen[r] = r;
    vector<Edge> Q(n), in(n, {-1,-1}), comp;
    deque<tuple<int, int, vector<Edge>>> cycs;
    rep(s, 0, n) {
        int u = s, qi = 0, w;
        while (seen[u] < 0) {
            if (!heap[u]) return {-1, {}};
            Edge e = heap[u]->top();
            heap[u]->delta -= e.w, pop(heap[u]);
            Q[qi] = e, path[qi++] = u, seen[u] = s;
            res += e.w, u = uf.find(e.a);
            if (seen[u] == s) {
                Node* cyc = 0;
                int end = qi, time = uf.time();
                do cyc = merge(cyc, heap[w = path[--qi]]);
                while (uf.join(u, w));
                u = uf.find(u), heap[u] = cyc, seen[u] = -1;
                cycs.push_front({u, time, {&Q[qi], &Q[end]}});
            }
        }
        rep(i, 0, qi) in[uf.find(Q[i].b)] = Q[i];
    }
}

```

```

for (auto& [u,t,comp] : cycs) { // restore sol (optional)
    uf.rollback(t);
    Edge inEdge = in[u];
    for (auto& e : comp) in[uf.find(e.b)] = e;
    in[uf.find(inEdge.b)] = inEdge;
}
rep(i, 0, n) par[i] = in[i].a;
return {res, par};}

```

TreeFlatten.h

Description: Subtree of a node is [in[node]...out[node]] The node itself is [in[node]...in[node]]

Time: ...

b731c2, 14 lines

```

vll in(N), out(N);    ll timer = -1;
void euler_tour(ll node, ll par) {
    in[node] = ++timer;
}

```

39e620, 59 lines

```
for(ll ch : gr[node]) {
    if(ch == par) continue;
    euler_tour(ch, node);
}
out[node] = timer;
}
// usage :
euler_tour(1, 0);
vll flat_tree(n);
L(i, 1, n) flat_tree[in[i]] = a[i];
sumSeg seg(flat_tree);
```

CentDecom.h
Description: ...
Time: ...

b47437, 35 lines

```
ll n,m,x,y,z,q,k,u,v,w;
vector<int> gr[N];
int sz[N];
int tot,done[N],cenpar[N];
```

```
void calc_sz(int node,int p){
    tot++; sz[node]=1;
    for(auto ch:gr[node]){
        if(ch==p||done[ch]) continue;
        calc_sz(ch,node); sz[node]+=sz[ch]; } }
```

```
int find_cen(int node,int p){ // find centroid
    for(auto ch:gr[node]){
        if(ch==p||done[ch]) continue;
        else if(sz[ch]>tot/2) return find_cen(ch,node); }
    return node; }
```

```
void decompose(int node,int p){ // find centroid of subtree
    tot=0; calc_sz(node,p);
    int cen=find_cen(node,p);
    cenpar[cen]=p; done[cen]=1;
    for(auto ch:gr[cen]){
        if(ch==p||done[ch]) continue;
        decompose(ch,cen); } }
```

```
vll cen_tree[N]; // centroid tree
```

```
int form_cen_tree(){ // form graph, return root(centroid)
    decompose(1,0); int root;
    L(i,1,n){
        if(cenpar[i]==0) root=i;
        if(cenpar[i]!=0){
            cen_tree[i].push_back(cenpar[i]);
            cen_tree[cenpar[i]].push_back(i); } }
    return root; }
```

6.8 Math

6.8.1 Number of Spanning Trees

Create an $N \times N$ matrix mat , and for each edge $a \rightarrow b \in G$, do $\text{mat}[a][b]--$, $\text{mat}[b][b]++$ (and $\text{mat}[b][a]--$, $\text{mat}[a][a]++$ if G is undirected). Remove the i th row and column and take the determinant; this yields the number of directed spanning trees rooted at i (if G is undirected, remove any row/column).

6.8.2 Erdős–Gallai theorem

A simple graph with node degrees $d_1 \geq \dots \geq d_n$ exists iff $d_1 + \dots + d_n$ is even and for every $k = 1 \dots n$,

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k).$$

Geometry (7)

7.1 Geometric primitives

Point.h

Description: Class to handle points in the plane. T can be e.g. double or long long. (Avoid int.)

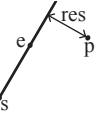
47ec0a, 28 lines

```
template<class T> int sgn(T x) { return (x > 0) - (x < 0); }
template<class T>
struct Point {
    typedef Point P;
    T x, y;
    explicit Point(T x=0, T y=0) : x(x), y(y) {}
    bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
    bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
    P operator+(P p) const { return P(x+p.x, y+p.y); }
    P operator-(P p) const { return P(x-p.x, y-p.y); }
    P operator*(T d) const { return P(x*d, y*d); }
    P operator/(T d) const { return P(x/d, y/d); }
    T dot(P p) const { return x*p.x + y*p.y; }
    T cross(P p) const { return x*p.y - y*p.x; }
    T cross(P a, P b) const { return (a-*this).cross(b-*this); }
    T dist2() const { return x*x + y*y; }
    double dist() const { return sqrt((double)dist2()); }
    // angle to x-axis in interval [-pi, pi]
    double angle() const { return atan2(y, x); }
    P unit() const { return *this/dist(); } // makes dist()==1
    P perp() const { return P(-y, x); } // rotates +90 degrees
    P normal() const { return perp().unit(); }
    // returns point rotated 'a' radians ccw around the origin
    P rotate(double a) const {
        return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); }
    friend ostream& operator<<(ostream& os, P p) {
        return os << "(" << p.x << ", " << p.y << ")"; }
};
```

lineDistance.h

Description:

Returns the signed distance between point p and the line containing points a and b. Positive value on left side and negative on right as seen from a towards b. a==b gives nan. P is supposed to be Point<T> or Point3D<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long. Using Point3D will always give a non-negative distance. For Point3D, call .dist on the result of the cross product.



f6bf6b, 4 lines

```
template<class P>
double lineDist(const P& a, const P& b, const P& p) {
    return (double) (b-a).cross(p-a)/(b-a).dist();
}
```

SegmentDistance.h

Description:

Returns the shortest distance between point p and the line segment from point s to e.



```
Usage: Point<double> a, b(2,2), p(1,1);
bool onSegment = segDist(a,b,p) < 1e-10;
"Point.h"
5c88f4, 6 lines

typedef Point<double> P;
double segDist(P& s, P& e, P& p) {
    if (s==e) return (p-s).dist();
    auto d = (e-s).dist2(), t = min(d,max(.0, (p-s).dot(e-s)));
    return ((p-s)*d-(e-s)*t).dist()/d;
}
```

SegmentIntersection.h

Description:

If a unique intersection point between the line segments going from s1 to e1 and from s2 to e2 exists then it is returned. If no intersection point exists an empty vector is returned. If infinitely many exist a vector with 2 elements is returned, containing the endpoints of the common line segment. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or long long.



```
Usage: vector<P> inter = segInter(s1,e1,s2,e2);
if (sz(inter)==1)
    cout << "segments intersect at " << inter[0] << endl;
"Point.h", "OnSegment.h"
9d57f2, 13 lines

template<class P> vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b),
        oc = a.cross(b, c), od = a.cross(b, d);
    // Checks if intersection is single non-endpoint point.
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa)};
    set<P> s;
    if (onSegment(c, d, a)) s.insert(a);
    if (onSegment(c, d, b)) s.insert(b);
    if (onSegment(a, b, c)) s.insert(c);
    if (onSegment(a, b, d)) s.insert(d);
    return {all(s)};
}
```

lineIntersection.h

Description:

If a unique intersection point of the lines going through s1,e1 and s2,e2 exists {1, point} is returned. If no intersection point exists {0, (0,0)} is returned and if infinitely many exists {-1, (0,0)} is returned. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or ll.



```
Usage: auto res = lineInter(s1,e1,s2,e2);
if (res.first == 1)
    cout << "intersection point at " << res.second << endl;
"Point.h"
a01f81, 8 lines

template<class P>
pair<int, P> lineInter(P s1, P e1, P s2, P e2) {
    auto d = (e1 - s1).cross(e2 - s2);
    if (d == 0) // if parallel
        return {-(s1.cross(e1, s2) == 0), P(0, 0)};
    auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
    return {1, (s1 * p + e1 * q) / d};
}
```

sideOf.h

Description:

Returns where p is as seen from s towards e. 1/0/-1 $\Leftrightarrow$ left/on line/right. If the optional argument eps is given 0 is returned if p is within distance eps from the line. P is supposed to be Point<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long.

Usage: bool left = sideOf(p1,p2,q)==1;

"Point.h"3af81c, 9 lines

```
template<class P>
int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }

template<class P>
int sideOf(const P& s, const P& e, const P& p, double eps) {
    auto a = (e-s).cross(p-s);
    double l = (e-s).dist()*eps;
    return (a > l) - (a < -l);
}
```

OnSegment.h

Description: Returns true iff p lies on the line segment from s to e. Use (segDist(s,e,p)<=epsilon) instead when using Point<double>.

"Point.h"c597e8, 3 lines

```
template<class P> bool onSegment(P s, P e, P p) {
    return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
}
```

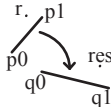
linearTransformation.h

Description:

Apply the linear transformation (translation, rotation and scaling) which takes line p0-p1 to line q0-q1 to point r.

"Point.h"03a306, 6 lines

```
typedef Point<double> P;
P linearTransformation(const P& p0, const P& p1,
    const P& q0, const P& q1, const P& r) {
    P dp = p1-p0, dq = q1-q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + P((r-p0).cross(num), (r-p0).dot(num))/dp.dist2();
}
```



Angle.h

Description: A class for ordering angles (as represented by int points and a number of rotations around the origin). Useful for rotational sweeping. Sometimes also represents points or vectors.

Usage: vector<Angle> v = {w[0], w[0].t360() ...}; // sorted

int j = 0; rep(i,0,n) { while (v[j] < v[i].t180()) ++j; }

// sweeps j such that (j-i) represents the number of positively oriented triangles with vertices at 0 and i

0f0602, 35 lines

struct Angle {
 int x, y;
 int t;
 Angle(int x, int y, int t=0) : x(x), y(y), t(t) {}
 Angle operator-(Angle b) const { return {x-b.x, y-b.y, t}; }
 int half() const {
 assert(x || y);
 return y < 0 || (y == 0 && x < 0);
 }
 Angle t90() const { return {-y, x, t + (half() && x >= 0)}; }
 Angle t180() const { return {-x, -y, t + half()}; }
 Angle t360() const { return {x, y, t + 1}; }
};
bool operator<(Angle a, Angle b) {
 // add a.dist2() and b.dist2() to also compare distances
 return make_tuple(a.t, a.half(), a.y * (1l)b.x) <
 make_tuple(b.t, b.half(), a.x * (1l)b.y);
}

// Given two points, this calculates the smallest angle between
// them, i.e., the angle that covers the defined line segment.
pair<Angle, Angle> segmentAngles(Angle a, Angle b) {
 if (b < a) swap(a, b);
 return (b < a.t180() ?
 make_pair(a, b) : make_pair(b, a.t360()));
}

Angle operator+(Angle a, Angle b) { // point a + vector b
 Angle r(a.x + b.x, a.y + b.y, a.t);
 if (a.t180() < r) r.t--;
 return r.t180() < a ? r.t360() : r;
}
Angle angleDiff(Angle a, Angle b) { // angle b - angle a
 int tu = b.t - a.t; a.t = b.t;
 return {a.x*b.x + a.y*b.y, a.x*b.y - a.y*b.x, tu - (b < a)};
}

7.2 Circles

CircleIntersection.h

Description: Computes the pair of points at which two circles intersect. Returns false in case of no intersection.

"Point.h"84d6d3, 11 lines

```
typedef Point<double> P;
bool circleInter(P a,P b,double r1,double r2,pair<P, P>* out) {
    if (a == b) { assert(r1 != r2); return false; }
    P vec = b - a;
    double d2 = vec.dist2(), sum = r1+r2, dif = r1-r2,
        p = (d2 + r1*r1 - r2*r2)/(d2*2), h2 = r1*r1 - p*p*d2;
    if (sum*sum < d2 || dif*dif > d2) return false;
    P mid = a + vec*p, per = vec.perp() * sqrt(fmax(0, h2) / d2);
    *out = {mid + per, mid - per};
    return true;
}
```

CircleTangents.h

Description: Finds the external tangents of two circles, or internal if r2 is negated. Can return 0, 1, or 2 tangents – 0 if one circle contains the other (or overlaps it, in the internal case, or if the circles are the same); 1 if the circles are tangent to each other (in which case .first = .second and the tangent line is perpendicular to the line between the centers). .first and .second give the tangency points at circle 1 and 2 respectively. To find the tangents of a circle with a point set r2 to 0.

"Point.h"b0153d, 13 lines

```
template<class P>
vector<pair<P, P>> tangents(P c1, double r1, P c2, double r2) {
    P d = c2 - c1;
    double dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 == 0 || h2 < 0) return {};
    vector<pair<P, P>> out;
    for (double sign : {-1, 1}) {
        P v = (d * dr + d.perp() * sqrt(h2) * sign) / d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    if (h2 == 0) out.pop_back();
    return out;
}
```

CirclePolygonIntersection.h

Description: Returns the area of the intersection of a circle with a ccw polygon.

Time: $\mathcal{O}(n)$

"../content/geometry/Point.h"19add1, 19 lines

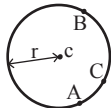
```
typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
    auto tri = [&](P p, P q) {
        auto r2 = r * r / 2;
        P d = q - p;
        auto a = d.dot(p)/d.dist2(), b = (p.dist2()-r*r)/d.dist2();
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a-sqrt(det)), t = min(1., -a+sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = q + d * (t-1);
```

```
    return arg(p,u) * r2 + u.cross(v)/2 + arg(v,q) * r2;
};
auto sum = 0.0;
rep(i,0,sz(ps))
    sum += tri(ps[i] - c, ps[(i + 1) % sz(ps)] - c);
return sum;
}
```

circumcircle.h

Description:

The circumcircle of a triangle is the circle intersecting all three vertices. ccRadius returns the radius of the circle going through points A, B and C and ccCenter returns the center of the same circle.



"Point.h"1caa3a, 9 lines

```
typedef Point<double> P;
double ccRadius(const P& A, const P& B, const P& C) {
    return (B-A).dist()*(C-B).dist()*(A-C).dist()/
        abs((B-A).cross(C-A))/2;
}
P ccCenter(const P& A, const P& B, const P& C) {
    P b = C-A, c = B-A;
    return A + (b*c.dist2()-c*b.dist2()).perp()/b.cross(c)/2;
}
```

MinimumEnclosingCircle.h

Description: Computes the minimum circle that encloses a set of points.

Time: expected $\mathcal{O}(n)$

"circumcircle.h"09dd0a, 17 lines

```
pair<P, double> mec(vector<P> ps) {
    shuffle(all(ps), mt19937(time(0)));
    P o = ps[0];
    double r = 0, EPS = 1 + 1e-8;
    rep(i,0,sz(ps)) if ((o - ps[i]).dist() > r * EPS) {
        o = ps[i], r = 0;
        rep(j,0,i) if ((o - ps[j]).dist() > r * EPS) {
            o = (ps[i] + ps[j]) / 2;
            r = (o - ps[i]).dist();
            rep(k,0,j) if ((o - ps[k]).dist() > r * EPS) {
                o = ccCenter(ps[i], ps[j], ps[k]);
                r = (o - ps[i]).dist();
            }
        }
    }
    return {o, r};
}
```

7.3 Polygons

InsidePolygon.h

Description: Returns true if p lies within the polygon. If strict is true, it returns false for points on the boundary. The algorithm uses products in intermediate steps so watch out for overflow.

Usage: vector<P> v = {P{4,4}, P{1,2}, P{2,1}};

bool in = inPolygon(v, P{3, 3}, false);

Time: $\mathcal{O}(n)$

"Point.h", "OnSegment.h", "SegmentDistance.h"2bf504, 11 lines

```
template<class P>
bool inPolygon(vector<P> &p, P a, bool strict = true) {
    int cnt = 0, n = sz(p);
    rep(i,0,n) {
        P q = p[(i + 1) % n];
        if (onSegment(p[i], q, a)) return !strict;
        //or: if (segDist(p[i], q, a) <= eps) return !strict;
        cnt ^= ((a.y<p[i].y) - (a.y<q.y)) * a.cross(p[i], q) > 0;
    }
    return cnt;
}
```

PolygonArea.h

Description: Returns twice the signed area of a polygon. Clockwise enumeration gives negative area. Watch out for overflow if using int as T!

```
"Point.h"
f12300, 6 lines

template<class T>
T polygonArea2(vector<Point<T>>& v) {
    T a = v.back().cross(v[0]);
    rep(i,0,sz(v)-1) a += v[i].cross(v[i+1]);
    return a;
}
```

PolygonCenter.h

Description: Returns the center of mass for a polygon.

Time: $\mathcal{O}(n)$

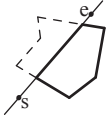
```
"Point.h"
9706dc, 9 lines

typedef Point<double> P;
P polygonCenter(const vector<P>& v) {
    P res(0, 0); double A = 0;
    for (int i = 0, j = sz(v) - 1; i < sz(v); j = i++) {
        res = res + (v[i] + v[j]) * v[j].cross(v[i]);
        A += v[j].cross(v[i]);
    }
    return res / A / 3;
}
```

PolygonCut.h

Description:
Returns a vector with the vertices of a polygon with everything to the left of the line going from s to e cut away.

Usage: vector<P> p = ...;
p = polygonCut(p, P(0,0), P(1,0));



```
"Point.h"
d07181, 13 lines

typedef Point<double> P;
vector<P> polygonCut(const vector<P>& poly, P s, P e) {
    vector<P> res;
    rep(i,0,sz(poly)) {
        P cur = poly[i], prev = i ? poly[i-1] : poly.back();
        auto a = s.cross(e, cur), b = s.cross(e, prev);
        if ((a < 0) != (b < 0))
            res.push_back(cur + (prev - cur) * (a / (a - b)));
        if (a < 0)
            res.push_back(cur);
    }
    return res;
}
```

ConvexHull.h

Description:
Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull.

Time: $\mathcal{O}(n \log n)$



```
"Point.h"
310954, 13 lines

typedef Point<ll> P;
vector<P> convexHull(vector<P> pts) {
    if (sz(pts) <= 1) return pts;
    sort(all(pts));
    vector<P> h(sz(pts)+1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(pts)))
        for (P p : pts) {
            while (t >= s + 2 && h[t-2].cross(h[t-1], p) <= 0) t--;
            h[t++] = p;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}
```

HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).

Time: $\mathcal{O}(n)$

```
"Point.h"
c571b8, 12 lines

typedef Point<ll> P;
array<P, 2> hullDiameter(vector<P> S) {
    int n = sz(S), j = n < 2 ? 0 : 1;
    pair<ll, array<P, 2>> res({0, {S[0], S[0]}});
    rep(i,0,j)
        for (; j = (j + 1) % n) {
            res = max(res, {(S[i] - S[j]).dist2(), {S[i], S[j]}});
            if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)
                break;
        }
    return res.second;
}
```

PointInsideHull.h

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included.

Time: $\mathcal{O}(\log N)$

```
"Point.h", "sideOf.h", "OnSegment.h"
71446b, 14 lines

typedef Point<ll> P;

bool inHull(const vector<P>& l, P p, bool strict = true) {
    int a = 1, b = sz(l) - 1, r = !strict;
    if (sz(l) < 3) return r && onSegment(l[0], l.back(), p);
    if (sideOf(l[0], l[a], l[b]) > 0) swap(a, b);
    if (sideOf(l[0], l[a], p) >= r || sideOf(l[0], l[b], p) <= -r)
        return false;
    while (abs(a - b) > 1) {
        int c = (a + b) / 2;
        (sideOf(l[0], l[c], p) > 0 ? b : a) = c;
    }
    return sgn(l[a].cross(l[b], p)) < r;
}
```

LineHullIntersection.h

Description: Line-convex polygon intersection. The polygon must be ccw and have no collinear points. lineHull(line, poly) returns a pair describing the intersection of a line with the polygon: $\bullet(-1, -1)$ if no collision, $\bullet(i, -1)$ if touching the corner i , $\bullet(i, i)$ if along side $(i, i + 1)$, $\bullet(i, j)$ if crossing sides $(i, i + 1)$ and $(j, j + 1)$. In the last case, if a corner i is crossed, this is treated as happening on side $(i, i + 1)$. The points are returned in the same order as the line hits the polygon. extrVertex returns the point of a hull with the max projection onto a line.

Time: $\mathcal{O}(\log n)$

```
"Point.h"
7cf45b, 39 lines

#define cmp(i, j) sgn(dir.perp().cross(poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i + 1, i) >= 0 && cmp(i, i - 1 + n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (extr(m)) return m;
        int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
    }
    return lo;
}

#define cmpl(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
```

```
int endB = extrVertex(poly, (b - a).perp());
if (cmpl(endA) < 0 || cmpl(endB) > 0)
    return {-1, -1};
array<int, 2> res;
rep(i,0,2) {
    int lo = endB, hi = endA, n = sz(poly);
    while ((lo + 1) % n != hi) {
        int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
        (cmpl(m) == cmpl(endB) ? lo : hi) = m;
    }
    res[i] = (lo + !cmpl(hi)) % n;
    swap(endA, endB);
}
if (res[0] == res[1]) return {res[0], -1};
if (!cmpl(res[0]) && !cmpl(res[1]))
    switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
        case 0: return {res[0], res[0]};
        case 2: return {res[1], res[1]};
    }
return res;
}
```

7.4 Misc. Point Set Problems

ClosestPair.h

Description: Finds the closest pair of points.

Time: $\mathcal{O}(n \log n)$

```
"Point.h"
ac41a6, 17 lines

typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
    assert(sz(v) > 1);
    set<P> S;
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d{1 + (ll)sqrt(ret.first), 0};
        while (v[j].y <= p.y - d.x) S.erase(v[j++]);
        auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
        for (; lo != hi; ++lo)
            ret = min(ret, {( *lo - p).dist2(), { *lo, p } });
        S.insert(p);
    }
    return ret.second;
}
```

7.5 3D

PolyhedronVolume.h

Description: Magic formula for the volume of a polyhedron. Faces should point outwards.

```
3058c3, 6 lines

template<class V, class L>
double signedPolyVolume(const V& p, const L& trilst) {
    double v = 0;
    for (auto i : trilst) v += p[i.a].cross(p[i.b]).dot(p[i.c]);
    return v / 6;
}
```

Point3D.h

Description: Class to handle points in 3D space. T can be e.g. double or long long.

```
8058ae, 32 lines

template<class T> struct Point3D {
    typedef Point3D P;
    typedef const P& R;
    T x, y, z;
    explicit Point3D(T x=0, T y=0, T z=0) : x(x), y(y), z(z) {}
    bool operator<(R p) const {
        return tie(x, y, z) < tie(p.x, p.y, p.z);
    }
```

```
bool operator==(R p) const {
    return tie(x, y, z) == tie(p.x, p.y, p.z); }
P operator+(R p) const { return P(x+p.x, y+p.y, z+p.z); }
P operator-(R p) const { return P(x-p.x, y-p.y, z-p.z); }
P operator*(T d) const { return P(x*d, y*d, z*d); }
P operator/(T d) const { return P(x/d, y/d, z/d); }
T dot(R p) const { return x*p.x + y*p.y + z*p.z; }
P cross(R p) const {
    return P(y*p.z - z*p.y, z*p.x - x*p.z, x*p.y - y*p.x);
}
T dist2() const { return x*x + y*y + z*z; }
double dist() const { return sqrt((double)dist2()); }
//Azimuthal angle (longitude) to x-axis in interval [-pi, pi]
double phi() const { return atan2(y, x); }
//Zenith angle (latitude) to the z-axis in interval [0, pi]
double theta() const { return atan2(sqrt(x*x+y*y),z); }
P unit() const { return *this/(T)dist(); } //makes dist()==1
//returns unit vector normal to *this and p
P normal(P p) const { return cross(p).unit(); }
//returns point rotated 'angle' radians ccw around axis
P rotate(double angle, P axis) const {
    double s = sin(angle), c = cos(angle); P u = axis.unit();
    return u*dot(u)*(1-c) + (*this)*c - cross(u)*s;
}
};
```

sphericalDistance.h

Description: Returns the shortest distance on the sphere with radius r adius between the points with azimuthal angles (longitude) f_1 (ϕ_1) and f_2 (ϕ_2) from x axis and zenith angles (latitude) t_1 (θ_1) and t_2 (θ_2) from z axis ($0 =$ north pole). All angles measured in radians. The algorithm starts by converting the spherical coordinates to cartesian coordinates so if that is what you have you can use only the two last rows. $dx*$ radius is then the difference between the two points in the x direction and $d*$ radius is the total distance between the points.

611f07, 8 lines

```
double sphericalDistance(double f1, double t1,
    double f2, double t2, double radius) {
    double dx = sin(t2)*cos(f2) - sin(t1)*cos(f1);
    double dy = sin(t2)*sin(f2) - sin(t1)*sin(f1);
    double dz = cos(t2) - cos(t1);
    double d = sqrt(dx*dx + dy*dy + dz*dz);
    return radius*2*asin(d/2);
}
```

Strings (8)

KMP.h

Time: $\mathcal{O}(n)$ 3ac13f, 29 lines

```
// for each position (0 based ***) of s, what is the best match
// of a suffix at that position with a prefix of s
vector<int> prefixFunction(const string& s){
    int n=s.size(); vector<int> pi(n); pi[0]=0;
    for(int i=1; i<n; ++i){
        int j = pi[i-1];
        while(j>0 and s[j] != s[i]) j = pi[j-1];
        if(s[j] == s[i]) j++;
        pi[i]=j;
    }
    return pi;
}
```

// if the automaton is built on string s , then :

// $aut[j][ch]$ = if i 'm scanning/constructing some string and rn the best match between $prefix_of_s$ and $suffix_of_MY_string$

sphericalDistance KMP Zfunc Manacher Trie Hashing

```
// is of LENGTH j, and the next scanned/put character is ch,
// then what would be the best matched LENGTH
vector<vector<int>> prefixAutomaton(const string& s){
    int m = s.size();
    vector<int> pi = prefixFunction(s);
    const int alph = 26;
    vector<vector<int>>> aut(m+1, vector<int>(alph));
    for(int j = 0; j <= m; j++){
        for(int c=0; c<alph; c++){
            if(j<m && s[j] == char('a'+c)) aut[j][c]= j+1;
            else if(j == 0) aut[j][c] = 0;
            else aut[j][c] = aut[pi[j-1]][c];
        }
    }
    return aut;
}
```

Zfunc.h

Description: $z[i]$ computes the length of the longest common prefix of $s[i:]$ and s , except $z[0] = 0$. (abacaba -> 0010301)

Time: $\mathcal{O}(n)$

f71b39, 12 lines

```
vi Z(const string& s) {
    vi z(sz(s));
    int l = -1, r = -1;
    rep(i,l,sz(s)) {
        z[i] = i >= r ? 0 : min(r - i, z[i - l]);
        while (i + z[i] < sz(s) && s[i + z[i]] == s[z[i]])
            z[i]++;
        if (i + z[i] > r)
            l = i, r = i + z[i];
    }
    return z;
}
```

Manacher.h

Description: For each position in a string, computes $p[0][i] =$ radius of longest even palindrome where central pair is $(i-1, i)$, $p[1][i] =$ radius of longest odd (center EXCLUDED).

Time: $\mathcal{O}(N)$

e7ad79, 13 lines

```
array<vi, 2> manacher(const string& s) {
    int n = sz(s);
    array<vi,2> p = {vi(n+1), vi(n)};
    rep(z,0,2) for (int i=0, l=0, r=0; i < n; i++) {
        int t = r-i+!z;
        if (i<r) p[z][i] = min(t, p[z][l+t]);
        int L = i - p[z][i], R = i + p[z][i] - !z;
        while (L>=1 && R+1<n && s[L-1] == s[R+1])
            p[z][i]++, L--, R++;
        if (R>r) l=L, r=R;
    }
    return p;
}
```

Trie.h

Description: ...

Time: ...

6db610, 70 lines

```
struct Trie{
    static const int ALPHABET=26;
    static const char BASE='a';
    struct Node{
        int next[ALPHABET];
        int ends_here_cnt,count; // passes this node
        Node() {
            fill(next,next+ALPHABET,-1);
            ends_here_cnt=0; count=0;
        }
    };
};
```

```
};
vector<Node> nodes;
Trie(){ nodes.emplace_back(); } // root node

void insert(const string& s){
    int cur=0;
    for(char ch:s){
        int c = ch-BASE;
        if(nodes[cur].next[c] == -1){
            nodes[cur].next[c]=nodes.size(); nodes.emplace_back();
        }
        cur = nodes[cur].next[c]; nodes[cur].count++;
    }
    nodes[cur].ends_here_cnt++;
}

bool erase(const string& s){
    if(!search(s)) return false;
    int cur=0;
    for(char ch:s){
        int c=ch-BASE;
        cur=nodes[cur].next[c]; nodes[cur].count--;
    }
    nodes[cur].ends_here_cnt--;
    return true;
}

bool search(const string& s)const{
    int cur=0;
    for(char ch:s){
        int c=ch-BASE;
        if(nodes[cur].next[c]==-1
            || nodes[nodes[cur].next[c]].count==0) return false;
        cur=nodes[cur].next[c];
    }
    return nodes[cur].ends_here_cnt>0;
}

bool starts_with(const string& prefix)const{
    int cur=0;
    for(char ch:prefix){
        int c=ch-BASE;
        if(nodes[cur].next[c]==-1
            ||nodes[nodes[cur].next[c]].count==0) return false;
        cur=nodes[cur].next[c];
    }
    return true;
}

int count_prefix(const string& prefix)const{
    int cur=0;
    for(char ch:prefix){
        int c=ch-BASE;
        if(nodes[cur].next[c]==-1
            ||nodes[nodes[cur].next[c]].count==0) return 0;
        cur=nodes[cur].next[c];
    }
    return nodes[cur].count;
}
};
```

Hashing.h

Description: Self-explanatory methods for string hashing.

2d2a67, 44 lines

```
// Arithmetic mod 2^64-1. 2x slower than mod 2^64 and more
// code, but works on evil test data (e.g. Thue-Morse, where
// ABBA... and BAAB... of length 2^10 hash the same mod 2^64).
// "typedef ull H;" instead if you think test data is random,
// or work mod 10^9+7 if the Birthday paradox is not a problem.
```

```
typedef uint64_t ull;
struct H {
    ull x; H(ull x=0) : x(x) {}
    H operator+(H o) { return x + o.x + (x + o.x < x); }
    H operator-(H o) { return *this + ~o.x; }
    H operator*(H o) { auto m = (__uint128_t)x * o.x;
        return H((ull)m) + (ull)(m >> 64); }
    ull get() const { return x + !~x; }
    bool operator==(H o) const { return get() == o.get(); }
    bool operator<(H o) const { return get() < o.get(); }
};

static const H C = (1ll)1e11+3; // (order ~ 3e9; random also ok)

struct HashInterval {
    vector<H> ha, pw;
    HashInterval(string& str) : ha(sz(str)+1), pw(ha) {
        pw[0] = 1;
        rep(i,0,sz(str))
            ha[i+1] = ha[i] * C + str[i],
            pw[i+1] = pw[i] * C;
    }
    H hashInterval(int a, int b) { // hash [a, b)
        return ha[b] - ha[a] * pw[b - a];
    }
};

vector<H> getHashes(string& str, int length) {
    if (sz(str) < length) return {};
    H h = 0, pw = 1;
    rep(i,0,length)
        h = h * C + str[i], pw = pw * C;
    vector<H> ret = {h};
    rep(i,length,sz(str)) {
        ret.push_back(h = h * C + str[i] - pw * str[i-length]);
    }
    return ret;
}

H hashString(string& s){H h{}; for(char c:s) h=h*C+c;return h;}
```

HashTree.h

Description: point updates in a string + hashing
Time: ...

```
1fd6d5, 44 lines

mt19937 rng(chrono::steady_clock::now().time_since_epoch()).
count();
const ll M = 991831889;
const ll C = uniform_int_distribution<ll>(0.1 * M, 0.9 * M)(rng
);

struct HashTree {
    struct Hash {
        ll hash = 0;
        ll mul = 1;
        Hash operator+(Hash other) {
            return {(hash * other.mul + other.hash) % M, mul *
                other.mul % M};
        }
    };
    int n; vector<Hash> tree;

    HashTree(string s) {
        n = 1;
        while (n < s.size()) n *= 2;
        tree.resize(2 * n);
        for (int i = 0; i < s.size(); i++) {
            tree[n + i] = {s[i], C};
        }
        for (int i = n - 1; i >= 1; i--) {
```

HashTree SuffixArray AhoCorasick Eertree

```
tree[i] = tree[2 * i] + tree[2 * i + 1];
    }
}

void set(int k, char x) {
    k += n;
    tree[k] = {x, C};
    for (k /= 2; k >= 1; k /= 2) {
        tree[k] = tree[2 * k] + tree[2 * k + 1];
    }
}

ll hash(int l, int r) {
    l += n;
    r += n;
    Hash left, right;
    while (l <= r) {
        if (l % 2 == 1) left = left + tree[l++];
        if (r % 2 == 0) right = tree[r--] + right;
        l /= 2; r /= 2;
    }
    return (left + right).hash;
}

};
```

SuffixArray.h

Description: Builds suffix array for a string. sa[i] is the starting index of the suffix which is *i*'th in the sorted suffix array. The returned vector is of size $n + 1$, and sa[0] = n. The lcp array contains longest common prefixes for neighbouring strings in the suffix array: lcp[i] = lcp(sa[i], sa[i-1]), lcp[0] = 0. The input string must not contain any nul chars.
Time: $\mathcal{O}(n \log n)$

```
3123fc, 22 lines

struct SuffixArray {
    vi sa, lcp;
    SuffixArray(string s, int lim=256) { // or vector<int>
        s.push_back(0); int n = sz(s), k = 0, a, b;
        vi x(all(s)), y(n), ws(max(n, lim));
        sa = lcp = y, iota(all(sa), 0);
        for (int j = 0, p = 0; p < n; j = max(1ll, j * 2), lim = p)
            {
                p = j, iota(all(y), n - j);
                rep(i,0,n) if (sa[i] >= j) y[p++] = sa[i] - j;
                fill(all(ws), 0);
                rep(i,0,n) ws[x[i]]++;
                rep(i,1,lim) ws[i] += ws[i - 1];
                for (int i = n; i--;) sa[-ws[x[y[i]]]] = y[i];
                swap(x, y), p = 1, x[sa[0]] = 0;
                rep(i,1,n) a = sa[i - 1], b = sa[i], x[b] =
                    (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
            }
        for (int i = 0, j; i < n - 1; lcp[x[i++]] = k)
            for (k && k--, j = sa[x[i] - 1];
                s[i + k] == s[j + k]; k++);
    }
};
```

AhoCorasick.h

Description: ...
Time: linear

```
29052a, 58 lines

struct Aho{
    static const int K=26; // alphabet size
    struct Vertex{
        int next[K],link,term_link; // suffix link, terminal link
        bool output; // true if this node is end of some pattern
        vector<int> ids; // pattern ids ending here
        int len;
        Vertex(){
            fill(begin(next), end(next), -1);
            link=-1; term_link=-1; output=false; len=0;
        }
    };
};
```

```
}
};
vector<Vertex> t = {Vertex()}; // root = state 0

void add_string(const string& s,int id){
    int v=0;
    for(char ch:s){
        int c=ch-'a';
        if(t[v].next[c] == -1){
            t[v].next[c] = t.size(); t.emplace_back();
        }
        v=t[v].next[c];
    }
    t[v].output=true; t[v].ids.push_back(id);
}
```

```
void build_automaton(){ // build the automaton T.T
    queue<int> q;
    t[0].link=0; t[0].term_link=0;
    for(int c=0; c<K; c++){
        int u=t[0].next[c];
        if(u!=-1){ t[u].link=0; q.push(u); t[u].len=1; }
        else t[0].next[c]=0;// missing edge from root loops to root
    }
    while(!q.empty()){
        int v=q.front(); q.pop();
        for(int c=0; c<K; c++){
            int u=t[v].next[c];
            if(u!=-1){ // compute suffix link
                t[u].link = t[t[v].link].next[c];
                t[u].len = t[v].len+1;
                q.push(u);
            }
            else t[v].next[c] = t[t[v].link].next[c];
        }
        // compute terminal link
        if(t[t[v].link].output) t[v].term_link = t[v].link;
        else t[v].term_link = t[t[v].link].term_link;
    }
}
```

```
void form_tree(){
    gr.resize(t.size());
    L(state,1,t.size()-1){
        gr[t[state].term_link].push_back(state);
    }
};
```

Eertree.h

Description: ...
Time: ...

sz - 2 // no. of distinct palindromes
no. of unique palindromes is linear !!! may use with hashing

use cnt during construction. After extend(i):
last = node representing the longest palindrome ending at pos i
pt.t[pt.last].cnt; // no. of pals ending here, smaller included

use oc after calc.
t[i].oc //no. of occurence of this exact pal[smaller excluded]
sum(t[i].oc) // total no. of palindromic substrings

```
struct PalindromicTree {
    struct node {
        int nxt[26]; // transitions (adding char to both ends)
        int len; // length of palindrome
        int st, en; // start and end index in original string
    };
};
```

```
int link; // suffix link (largest palindromic suffix)
ll cnt; // no. of palindromic suffixes ending here
ll oc; // no. of occurrences of this palindrome
};
string s;
vector<node> t;
int sz, last;
PalindromicTree() {}
PalindromicTree(string _s) {
    s = _s;
    int n = s.size();
    t.clear();
    t.resize(n + 9);
    sz = 2, last = 2;
    t[1].len = -1, t[1].link = 1;
    t[2].len = 0, t[2].link = 1;
}
int extend(int pos){//returns 1 if it creates a new palindrome
    int cur = last, curlen = 0;
    int ch = s[pos] - 'a';
    while (1) {
        curlen = t[cur].len;
        if(pos-1-curlen >= 0 && s[pos-1-curlen] == s[pos]) break;
        cur = t[cur].link;
    }
    if (t[cur].nxt[ch]) {
        last = t[cur].nxt[ch];
        t[last].oc++;
        return 0;
    }
    sz++;
    last = sz;
    t[sz].oc = 1;
    t[sz].len = t[cur].len + 2;
    t[cur].nxt[ch] = sz;
    t[sz].en = pos;
    t[sz].st = pos - t[sz].len + 1;
    if (t[sz].len == 1) {
        t[sz].link = 2;
        t[sz].cnt = 1;
        return 1;
    }
    while (1) {
        cur = t[cur].link;
        curlen = t[cur].len;
        if (pos-1-curlen >= 0 && s[pos-1-curlen] == s[pos]) {
            t[sz].link = t[cur].nxt[ch];
            break;
        }
    }
    t[sz].cnt = 1 + t[t[sz].link].cnt;
    return 1;
}
void calc_occurrences() {
    for (int i = sz; i >= 3; i--) t[t[i].link].oc += t[i].oc;
}
};
```

SuffixAutomaton.h

Description: ...
Time: ...

b94687, 70 lines

/*
 Every substring can be formed with transitions from root.
 Each node represents some substrings, longest being of
 length maxlen, shortest = suflink.maxlen + 1
 so, no. of unique substrings = sum of all (maxlen - minlen + 1)
*/

```
struct SAM{
    struct State{
        int link, maxlen;
        map<char,int> next;
        State(){ link=-1; maxlen=0; }
    };

    vector<State> st;
    vector<vector<int>> gr;
    vector<ll> koybar;
    vector<bool> was_terminal;
    int last,sz;
    ll total; // number of unique substrings

    SAM(int n){
        st.resize(2*n); gr.resize(2*n); koybar.resize(2*n);
        was_terminal.resize(2*n);
        st[0]=State(); sz=1; last=0; total=0;
    }

    void extend(char c){
        int cur = sz++;
        st[cur].maxlen = st[last].maxlen+1;
        int p=last;
        while(p!=-1 && !st[p].next.count(c)){
            st[p].next[c]=cur; p=st[p].link;
        }
        if(p == -1){ st[cur].link=0; }
        else{
            int q = st[p].next[c];
            if(st[p].maxlen+1 == st[q].maxlen){ st[cur].link=q; }
            else{
                int clone=sz++;
                st[clone] = st[q];
                st[clone].maxlen = st[p].maxlen + 1;
                while(p!=-1 && st[p].next[c] == q){
                    st[p].next[c]=clone; p=st[p].link;
                }
                st[q].link = st[cur].link = clone;
            }
        }
        last=cur; was_terminal[last]=true;
        total+=st[cur].maxlen-st[st[cur].link].maxlen;
    }

    void build_tree(){ // USE (sam.sz-1) when you need to visit all states
        L(1,1,sz-1) gr[st[i].link].push_back(i);
    }

    void guno(){ // count repetitions in the string for each node
        int cur=last;
        L(1,1,sz-1) koybar[i] = was_terminal[i];
        function<void(int)> dfs=[&](int node){
            for(int ch:gr[node])
                { dfs(ch); koybar[node] += koybar[ch]; }
        };
        dfs(0);
    }

    ll koyta(int i){ return st[i].maxlen-st[st[i].link].maxlen; }

    // after adding the entire string, if you need terminal states
    :
    // start from last, go back by suffix links. all visited
    states are terminal states
};
```

MinRotation.h

Description: Finds the lexicographically smallest rotation of a string.
Usage: rotate(v.begin(), v.begin()+minRotation(v), v.end());
Time: O(N)

d07a42, 8 lines

```
int minRotation(string s) {
    int a=0, N=sz(s); s += s;
    rep(b,0,N) rep(k,0,N) {
        if (a+k == b || s[a+k] < s[b+k]) {b += max(0, k-1); break;}
        if (s[a+k] > s[b+k]) { a = b; break; }
    }
    return a;
}
```

SuffixTree.h

Description: Ukkonen's algorithm for online suffix tree construction. Each node contains indices [l, r] into the string, and a list of child nodes. Suffixes are given by traversals of this tree, joining [l, r] substrings. The root is 0 (has l = -1, r = 0), non-existent children are -1. To get a complete tree, append a dummy symbol – otherwise it may contain an incomplete path (still useful for substring matching, though).

Time: O(26N)

aae0b8, 47 lines

```
struct SuffixTree {
    enum { N = 200010, ALPHA = 26 }; // N ~ 2*maxlen+10
    int toi(char c) { return c - 'a'; }
    string a; // v = cur node, q = cur position
    int t[N][ALPHA], l[N], r[N], p[N], s[N], v=0, q=0, m=2;
    void ukkadd(int i, int c) { suff:
        if (r[v]<=q) {
            if (t[v][c]==-1) { t[v][c]=m; l[m]=i;
                p[m+]=v; v=s[v]; q=r[v]; goto suff; }
            v=t[v][c]; q=l[v];
        }
        if (q==-1 || c==toi(a[q])) q++; else {
            l[m+]=i; p[m+]=m; l[m]=l[v]; r[m]=q;
            p[m]=p[v]; t[m][c]=m+1; t[m][toi(a[q])]=v;
            l[v]=q; p[v]=m; t[p[m]][toi(a[l[m]])]=m;
            v=s[p[m]]; q=l[m];
            while (q<r[m]) { v=t[v][toi(a[q])]; q+=r[v]-l[v]; }
            if (q==r[m]) s[m]=v; else s[m]=m+2;
            q=r[v]-(q-r[m]); m+=2; goto suff;
        }
    }
    SuffixTree(string a) : a(a) {
        fill(r,r+N,sz(a));
        memset(s, 0, sizeof s);
        memset(t, -1, sizeof t);
        fill(t[1],t[1]+ALPHA,0);
        s[0] = 1; l[0] = 1[l] = -1; r[0] = r[1] = p[0] = p[1] = 0;
        rep(i,0,sz(a)) ukkadd(i, toi(a[i]));
    }
    // example: find longest common substring (uses ALPHA = 28)
    pii best;
    int lcs(int node, int il, int i2, int olen) {
        if (l[node] <= il && il < r[node]) return 1;
        if (l[node] <= i2 && i2 < r[node]) return 2;
        int mask = 0, len = node ? olen + (r[node] - l[node]) : 0;
        rep(c,0,ALPHA) if (t[node][c] != -1)
            mask |= lcs(t[node][c], il, i2, len);
        if (mask == 3)
            best = max(best, {len, r[node] - len});
        return mask;
    }
    static pii LCS(string s, string t) {
        SuffixTree st(s + (char)('z' + 1) + t + (char)('z' + 2));
        st.lcs(0, sz(s), sz(s) + 1 + sz(t), 0);
        return st.best;
    }
};
```

BitTrie.h	
Description: ...	
Time: ...	01eb16, 53 lines
<pre>struct BitTrie{ struct Node{ int child[2],cnt;//trie[0]=total numbers inserted in trie Node(){ child[0] = child[1] = -1; cnt=0; } }; vector<Node> trie; int BITS; BitTrie(int maxBits = 30){ // 30 for numbers up to 1e9 BITS = maxBits; trie.push_back(Node()); // root }</pre>	

```
void insert(int x){
    int node=0;
    trie[node].cnt++;
    for(int b=BITS; b>=0; b--){
        int bit = (x>>b)&1;
        if(trie[node].child[bit] == -1){
            trie[node].child[bit] = trie.size();
            trie.push_back(Node());
        }
        node = trie[node].child[bit]; trie[node].cnt++;
    }

    void erase(int x){
        int node=0;
        trie[node].cnt--;
        for(int b=BITS; b>=0; b--){
            int bit = (x>>b)&1;
            int nxt=trie[node].child[bit];
            node=nxt; trie[node].cnt--;
        }
    }

    ll max_xor(ll x){ // find y from this trie that maximizes x^y
        if(trie[0].cnt == 0) return 0; // empty trie
        ll cur=0, y=0;
        R(bit, BITS, 0){
            int nextbit = getBit(x,bit); // at root, nextbit=30
            if(trie[cur].child[!nextbit] != -1
                and trie[trie[cur].child[!nextbit]].cnt){
                cur = trie[cur].child[!nextbit];
                if(!nextbit) y = setbit(y,bit);
            }
            else{ // has to exist, since tire is not empty
                cur = trie[cur].child[nextbit];
                if(nextbit) y = setbit(y,bit);
            }
        }
        return x^y;
    }
};
```

Various (9)

9.1 Misc. algorithms

TernarySearch.h
Description: Find the smallest i in $[a,b]$ that maximizes $f(i)$, assuming that $f(a) < \dots < f(i) \geq \dots \geq f(b)$. To reverse which of the sides allows non-strict inequalities, change the $<$ marked with (A) to $\leq$, and reverse the loop at (B). To minimize f , change it to $>$, also at (B).
Usage: int ind = ternSearch(0,n-1,[&](int i){return a[i];});

Time: $\mathcal{O}(\log(b-a))$	9155b4, 11 lines
<pre>template<class F> int ternSearch(int a, int b, F f) { assert(a <= b); while (b - a >= 5) { int mid = (a + b) / 2; if (f(mid) < f(mid+1)) a = mid; // (A) else b = mid+1; } rep(i,a+1,b+1) if (f(a) < f(i)) a = i; // (B) return a; }</pre>	

LIS.h	
Description: Compute indices for the longest increasing subsequence.	
Time: $\mathcal{O}(N\log N)$	2932a0, 17 lines
<pre>template<class I> vi lis(const vector<I>& S) { if (S.empty()) return {}; vi prev(sz(S)); vi prev(sz(S)); typedef pair<I, int> p; vector<p> res; rep(i,0,sz(S)) { // change 0 -> i for longest non-decreasing subsequence auto it = lower_bound(all(res), p{S[i], 0}); if (it == res.end()) res.emplace_back(), it = res.end()-1; *it = {S[i], i}; prev[i] = it == res.begin() ? 0 : (it-1)->second; } int L = sz(res), cur = res.back().second; vi ans(L); while (L--) ans[L] = cur, cur = prev[cur]; return ans; }</pre>	

Diophantine.h	
Description: ...	
Time: ...	ac879d, 23 lines
<pre>// Extended Euclidean Algorithm ll ext_gcd(ll a,ll b,ll& x,ll& y){ if(b==0){ x=1,y=0; return a; } ll x1,y1,g=ext_gcd(b,a%b,x1,y1); x=y1; y=x1-(a/b)*y1; return g; } // Solve ax + by = c // general solution : x + k*b/g, y - k*a/g bool solve_diophantine(ll a,ll b,ll c,ll& x,ll& y,ll& g){ g=ext_gcd(abs(a),abs(b),x,y); if(c%g!=0) return false; x*=c/g; y*=c/g; if(a<0)x=-x; if(b<0)y=-y; return true; }</pre>	

```
bool solve_dioph_non_neg(ll a,ll b,ll c,ll& x,ll& y,ll& g){
    if(!solve_diophantine(a,b,c,x,y,g)) return false;
    ll dx=(b/g),dy=(a/g);
    if(x<0){
        ll k=(-x+abs(dx)-1)/abs(dx); // ceil
        if(dx<0)k=-k; x+=k*dx; y-=k*dy; }
    if(y<0){
        ll k=(-y+abs(dy)-1)/abs(dy);
        if(dy<0)k=-k; y+=k*dy; x-=k*dx; }
    return x>=0 and y>=0; }
```

9.2 Dynamic programming

SOSDP.h	
Description: ...	
Time: ...	1c9cc3, 31 lines

<pre>// for problems like sum/count involving submaks/supermask, // we use sos dp to avoid overcounting. // x y = x : y is submask of x // x&y = x : y is supermask of x // x&y = 0 : y is submask of ~x // k = koyta bit max void fwd1(vll& dp, int k){ // dp[x] = cnt of submask of x int N = 1ll<<k; L(bit,0,k-1){ L(mask,0,N-1){ if(isSet(mask,bit)) dp[mask]+=dp[resetbit(mask,bit)]; } } } void bak1(vll& dp, int k){ // return from submask count to mask count int N = 1ll<<k; R(bit,19,0){ R(mask,N-1,0){ if(isSet(mask,bit)) dp[mask]==dp[resetbit(mask,bit)]; } } } void fwd2(vll& dp, int k){ // dp[x] = cnt of supermask of x int N = 1ll<<k; L(bit,0,19){ R(mask,N-1,0){ if(isSet(mask,bit)) dp[resetbit(mask,bit)]+=dp[mask]; } } } void bak2(vll& dp, int k){ // return from supermask count to mask count int N = 1ll<<k; R(bit,19,0){ L(mask,0,N-1){ if(isSet(mask,bit)) dp[resetbit(mask,bit)]-=dp[mask]; } } }</pre>	
---	--

DigitDP.h	
Description: ...	
Time: ...	ea364a, 26 lines
<pre>ll n,m,k,q; ll a[N]; ll dp[][11][2][2]; // put max len of string + 1</pre>	

```
bool is_valid(){ return 1; } // for even sum, return sum % 2 == 0;

ll cnt(string s,ll pos,ll prev,bool started,bool tight){
    if(pos==Size(s)) return is_valid(); // number of such numbers = 1
    if(dp[pos][prev][started][tight]!=-1) return dp[pos][prev][started][tight];
    ll ans=0;
    ll digMax=tight?s[pos]-'0':9;
    for(ll dig=0;dig<digMax;dig++){
        // ***
        if(started and dig==prev) continue; // no repitition
        bool startedNext=started?1:dig!=0; // started || (dig != 0)
        bool tightNext=tight?(dig==digMax):0; // tight && (dig == digMax)
        ans+=cnt(s,pos+1,dig,startedNext,tightNext);
    }
    return dp[pos][prev][started][tight]=ans; }
```

```
// usage:
string s1 = to_string(n1 - 1);
memset(dp, -1, sizeof(dp));
ll ans1; if(n1-1 < 0) ans1 = 0;
// 10 is a dummy value for prev.
else ans1 = cnt(s1, 0, 10, 0, 1);
// similarly, find ans2, and output : ans2-ans1
```

DivideAndConquerDP.h

Description: the dnc sol for cses subarray squares

Time: k*n*log(n) here 044ba1, 36 lines

```
/*
dnc dp can be applid for
dp[i] = min( dp[j] + cost(j+1..i) ) for all j<i
if opt(i) <= opt(i-1) is satisfied
where opt(i) = smallest j among all the best j for i
this may or may not be obvious
[if you cant prove, try bruteforcing too see if it holds]
one (not only) case when this is true is:
cost function satisfies quadrangle inequality:
cost(a..c) + cost(b..d) <= cost(a..d) + cost(b..c)
where a <= b <= c <= d
the cost function (pref[i]-pref[j])^2 satisfies this
this runs in nlogn for each k. so k*nlogn total
*/
const int MXN = 3005, inf = 2e18;
ll pref[MXN], dp[MXN][MXN]; //dp[i][k]=divide upto i int k segs
ll cost(int l, int r) { // change
    int sum = pref[r] - pref[l-1]; return sum*sum;
}
void dnc(int k, int l, int r, int optl, int optr) {
    if (l > r) return;
    int mid = (l + r) / 2;
    pair<int, int> best = {inf, 1};
    for (int j = optl; j <= min(mid - 1, optr); j++) {
        best = min(best, make_pair(dp[j][k-1] + cost(j+1, mid),
                                   j));
    }
    dp[mid][k] = best.first;
    int opt = best.second;
    dnc(k, l, mid - 1, optl, opt);
    dnc(k, mid + 1, r, opt, optr);
}
usage:
dp[0][0] = 0;
for (int i = 1; i <= n; i++) dp[i][0] = inf;
for (int k = 1; k <= _K; k++) dnc(k, 1, n, 0, n - 1);
// [optl, optr] = possible values of j. starts with [0,n-1]
```

submasks.h

Description: ...

Time: ... b0dc54, 2 lines

```
for (int mask = 0; mask < (1<<n); mask++)
    for(int i = mask; i; i = (i-1) & mask)
```

int128.h

Description: ...

Time: ... 6125c0, 14 lines

```
using i128=__int128_t;
i128 read(){
    string s; cin>>s; bool neg=false; size_t i=0;
    if(s[0]=='-'){ neg=true; i=1; } i128 mag=0;
    for (;i<s.size();++i){ mag=mag*10+(unsigned)(s[i]-'0'); }
    return neg?-mag:mag; }
void print(i128 x){
    if(x==0){cout<<0<<nl;return;}
    bool neg=x<0; if(neg)x=-x;
    string s;while(x){ int d=(int)(x%10);
        s.push_back(char('0'+d)); x/=10; }
    if(neg)s.push_back('-'); reverse(s.begin(),s.end());
    cout<<s<<nl; }
bool cmp(i128 x,i128 y){ return x>y; } // for sorting
```

FloorCeilDiv.h

Description: ...

Time: ... d77b32, 10 lines

```
ll floor_div(ll a, ll b) {
    if (b < 0) a = -a, b = -b;
    if (a >= 0) return a / b;
    return (a - b + 1) / b;
}
ll ceil_div(ll a, ll b) {
    if (b < 0) a = -a, b = -b;
    if (a >= 0) return (a + b - 1) / b;
    return a / b;
}
```

XorBasis.h

Description: ...

Time: ... 5be624, 59 lines

```
struct XorBasis {
    static const int B = 60; // for 64-bit numbers
    long long b[B]; int sz;
    XorBasis() {
        memset(b, 0, sizeof b); sz = 0;
    }
    // Insert number into basis
    void insert(long long x) {
        for (int i = B-1; i >= 0; i--) {
            if (!(x >> i & 1)) continue;
            if (b[i]) x ^= b[i]; // eliminate bit
            else { // new basis vector
                b[i] = x; sz++; return;
            }
        }
        // Check if some subset has xor = x
        bool can(long long x) {
            for(int i = B-1; i>=0; i--) x = min(x, x ^ b[i]);
            return x == 0;
        }
        // Maximum possible xor with optional start value
        long long max_xor(long long x = 0) {
            for(int i = B-1; i>=0; i--) x = max(x, x ^ b[i]);
            return x;
        }
        // k-th smallest subset xor (1-indexed, 1st = 0)
        long long kth(long long k) {
            if (k < 1 || k > (1LL << sz)) return -1;
            long long x = 0, cnt = 1LL << sz;
            for (int i = B-1; i >= 0; i--) if (b[i]) {
                if (k > cnt/2) {
                    if (!(x >> i & 1)) x ^= b[i]; k -= cnt/2;
                } else {
                    if (x >> i & 1) x ^= b[i];
                }
                cnt /= 2;
            }
            return x;
        }
        // Count how many subset xors are < x
        long long count_lt(long long x) {
            if (x < 0) return 0;
            long long ans = 0, cnt = 1LL << sz;
            long long mask = 0;
            for (int i = B-1; i >= 0; i--) {
                if (b[i]) {
                    if (x >> i & 1) {
                        ans += cnt/2;
                        if (!(mask >> i & 1)) mask ^= b[i];
                    } else if (mask >> i & 1) {
                        mask ^= b[i];
                    }
                }
            }
        }
    }
}
```

```
        cnt /= 2;
    }
    else if ((x >> i & 1) != (mask >> i & 1)) {
        return (x >> i & 1) ? ans + cnt : ans;
    }
}
return ans;
};
```

SubsetSumSqrt.h

Description: ...

Time: ... f9ccce, 28 lines

```
main:
// Sum of elements <= N implies that every element is <= N
vector<int> freq(N + 1, 0);
for (int i = 0; i < N; i++) {
    int x; cin >> x; freq[x]++;
}
vector<pair<int, int>> compressed;
for (int i = 1; i <= N; i++) {
    if (freq[i] > 0) compressed.emplace_back(i, freq[i]);
}
vector<int> dp(N + 1, 0); dp[0] = 1;
for (const auto &[w, k] : compressed) {
    vector<int> ndp = dp;
    for (int p = 0; p < w; p++) {
        int sum = 0;
        for(int mult = p, cnt = 0; mult<=N; mult += w, cnt++){
            if (cnt > k) {
                sum -= dp[mult - w * cnt]; cnt--;
            }
            if (sum > 0) ndp[mult] = 1; sum += dp[mult];
        }
        swap(dp, ndp);
    }
}
cout << "Possible subset sums are:\n";
for (int i = 0; i <= N; i++) {
    if (dp[i] > 0) cout << i << " ";
}
}
```